

Chapter 3

Transport in electronic systems

3.1 Introduction

The study of electronic transport phenomena in solids started several decades ago and is a well-studied problem by now. Transport evidently involves electronic bias in the form of including a battery in the circuit, thereby making a current flow. However, a battery can be replaced by a thermal gradient, which also results in the manifestation of current–voltage characteristics of the circuit.

Developing a formalism for the electronic transport in the presence of a bias necessitates inclusion of two important aspects, namely a non-equilibrium description, and a full quantum mechanical treatment of the system. A real system in general includes a very large number of particles, which is intractable in terms of computational logistics in any foreseeable future. Thus treating only a few particles (at the most of the order of 100) is manageable using exact techniques, and for numbers larger than that, one has to resort to controlled approximation methods. To treat systems out of equilibrium (but not too far from it), the formalism of non-equilibrium Green’s functions is popular (interested readers may see references [1, 2]), since the method retains important correlations in space and time that are very important in the study of transport properties.

Another important technique to address transport is via solving the Boltzmann transport equation (BTE) [3], where the time evolution of the distribution function that involves the positions and the momenta of the constituent particles that enter through the local density and the temperature of the system. The prime importance of this equation is to account for non-equilibrium phenomena, and hence discuss transport beyond the linear regime.

Yet another, and most widely used technique for exploring quantum transport is the Kubo formula, which describes conduction using linear transport theory. The method involves computing the matrix elements of the current operator between the eigenfunctions of the Hamiltonian. A standout advantage of this method lies in the ability to solve the system both in real and momentum spaces, and hence disorder and imperfections, which are indispensable in real materials, can be taken into account. We defer detailed discussion of this formalism for now, and come back to it later.

Transport in lower dimensions, or relevant to nanoscale and mesoscale devices is a subject of prolonged research interest. Depending on certain critical length scales, the transport may either be semi-classical, quantum, or an intermediate between the two. The last one is the most interesting scenario, since the effects of decoherence and dissipation have crucial roles on the transport properties, and quantum effects dominate the energy spectra. We address some of these in the following.

Mesoscopic physics is a relatively new branch of condensed matter physics. It deals with systems having dimensions that are intermediate between the microscopic (a few angstroms) and the macroscopic length (of the order of a micron) scale. A conductor of length L appears to be ohmic if $L \ll \lambda_{\text{db}}$ (λ_{db} : de Broglie wavelength), however, it is termed as non-ohmic or mesoscopic if $L \ll \lambda_{\phi}$ (λ_{ϕ} : phase coherence length, over which the electrons retain phase coherence). This regime is governed by the Landauer formula for conductance [4], namely,

$$G = \left(\frac{e^2}{h} \right) T$$

where T is the probability of an electron to be transmitted through the conductor and has been summed over all the channels or modes. The constant, $\frac{e^2}{h} = [(25.8)k\Omega]^{-1}$ represents the quantum of conductance corresponding to a single transverse mode in a conductor.

Several typical characteristic length scales are briefly introduced to define and characterize different transport regimes [1, 5, 6]. They are:

(a) Fermi wavelength (λ_F): It is the de Broglie wavelength (λ_{db}) of the electrons near the Fermi energy.

(b) Mean free path (l_e): It is the distance that an electron travels before its initial momentum is destroyed, and describes the elastic scattering caused by the random impurity potential.

(c) Phase coherent length (l_ϕ): It is the distance that an electron travels before it loses its initial phase due to scattering from other electrons or lattice vibrations.

(d) Localization length ξ : It is an asymptotic property of the conductance as a function of the linear sample size and can be expressed by the relation, $G = G_0 e^{-2L/\xi}$, where G_0 is the conductance without the disorder.

In terms of the above length scales, one can define three distinct regimes for the mesoscopic transport. These are:

(i) Ballistic regime ($\lambda_{db} < L \ll l_e, l_\phi$): In this regime, the electrons propagate through the mesoscopic sample without any elastic or phase breaking scattering in the bulk. The scattering occurs at the contact region separating the leads and the conductor and the impurity scattering can be neglected.

(ii) Diffusive regime ($\lambda_{db} \ll l_e \ll L < \xi$): In this regime, a few impurity atoms or a small concentration of disorder exist in the conducting sample and the electrons traverse coherently. Scattering is sample specific and one observes the full statistical fluctuations which are not suppressed by self-averaging.

(iii) Insulating regime ($\xi < L$): It's also called localized regime in the presence of strong scattering limit where the interference can completely cease the transport inside the disordered mesoscopic conductor.

Many novel phenomena exist that are intrinsic to mesoscopic systems, such as the integer quantum Hall effect, where the Hall resistance is quantized in units of h/e^2 , Aharonov–Bohm (AB) oscillations [7] in the conductance spectrum of mesoscopic rings with flux period h/e and ‘the dissipationless persistent currents’ flowing in such systems in the normal (non-superconducting) resistive states, etc. Since then, attention on the physics governing mesoscopic transport has been growing rapidly, and a wide range of new physical concepts have been realized. Some of them are: mesoscopic resistors in series that do not obey standard resistance addition rules [1, 2], the conductance of very narrow constrictions being quantized [8, 9], the conductance of disordered systems shows sample specific reproducible fluctuations with universal amplitudes, the weak localization phenomena [10, 11], etc.

Over the last two decades, it is possible to experimentally realize dimensions, $L < l_\phi$, using modern electronic material fabrication techniques, such as molecular beam epitaxy (MBE), lithography technique, scanning tunnelling microscope (STM), etc [12]. They are denoted as nanostructures. Familiar examples are GaAs/GaAlAs semiconductor heterostructures, which are formed by two semiconducting samples with different doping levels. At the interface, a thin layer of charges, called a two-dimensional electron gas is realized (2DEG). These heterostructures yield mobilities of the order of $10^6 \text{cm}^2/(\text{V}\cdot\text{s})$ that correspond to flow of electrons having a mean free path of about $10 \mu\text{m}$.

We shall not prolong this introduction on the transport phenomena at the mesoscale or nanoscale beyond this point, as that forms a subject of intense research interest on its own. Rather, we shall

concentrate on transport in two-dimensional electronic systems in the presence of a perpendicular magnetic field. This is known as the Hall effect, which the reader should be familiar with at the beginning phase of his/her undergraduate studies. We shall mainly discuss the quantum version of it, which, for obvious reasons is known as the quantum version of the Hall effect (QHE). QHE laid the foundation of studying condensed matter physics from a whole new perspective, and necessitated deeper introspection to connect the emergent quantum phenomena to the fundamental description of materials.

3.2 Quantum Hall effect

The date of discovery of the QHE is known pretty accurately. It occurred in the night between 4th and 5th February, 1980 at 2:00 AM in the morning at the high magnetic lab in Grenoble, France (see figure 3.1). There was ongoing research on the transport properties of silicon field-effect transistors (FETs). The main motive was to improve the mobility of these FET devices. They were provided by Dorda and Pepper which allowed direct measurement of the resistivity tensor. The system is a highly degenerate two-dimensional electron gas contained in the inversion layer of a metal-oxide semiconductor field-effect transistor (MOSFET) operated at low temperatures and strong magnetic fields. The original notes appear in figure 3.1, where it is clearly stated that the Hall resistivity involves universal constants and hence signals towards the involvement of a very fundamental phenomenon.

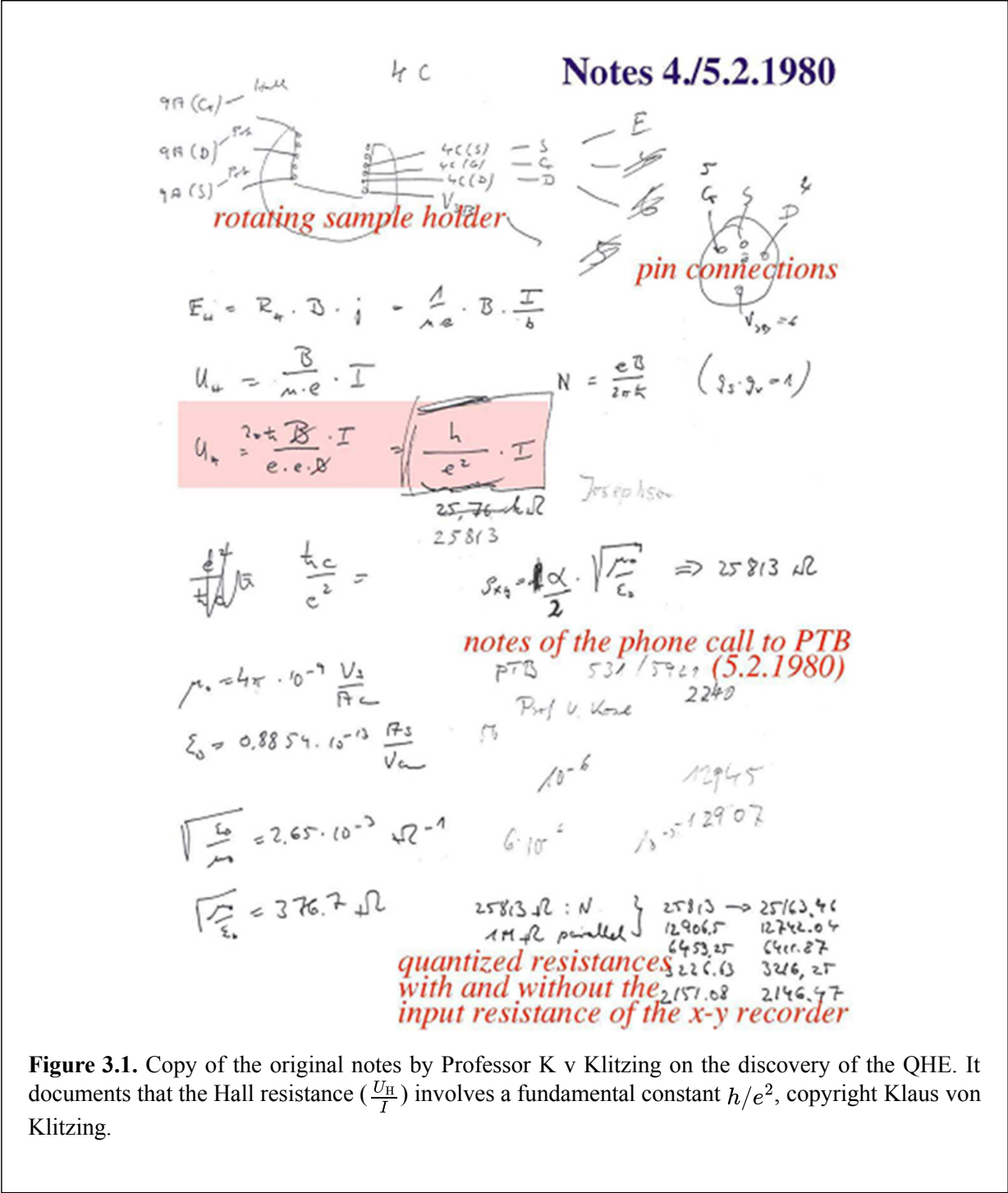


Figure 3.1. Copy of the original notes by Professor K v Klitzing on the discovery of the QHE. It documents that the Hall resistance ($\frac{U_H}{I}$) involves a fundamental constant h/e^2 , copyright Klaus von Klitzing.

In the classical version of the phenomenon discovered by E Hall in 1879, just over a hundred years before the discovery of its quantum analogue, one may consider a sample with a planar geometry so as to restrict the carriers to move in a two-dimensional (2D) plane. Next, turn on a bias voltage so that a current flows in one of the longitudinal directions and a strong magnetic field perpendicular to the plane of the gas (see figure 3.2). Because of the Lorentz force, the carriers drift towards a direction transverse to the direction of the current flowing in the sample. At equilibrium, a voltage develops in the transverse

direction, which is known as the Hall voltage. The Hall resistivity, R defined as the Hall voltage divided by the longitudinal current is found to linearly depend on the magnetic field, B and inversely on the carrier density, n through $R = \frac{B}{nq}$ (q is the charge). A related and possibly more familiar quantity is the Hall coefficient, denoted by, $R_H = R/B$ which, via its sign yields information on the type of the majority carriers, that is, whether they are electrons or holes.

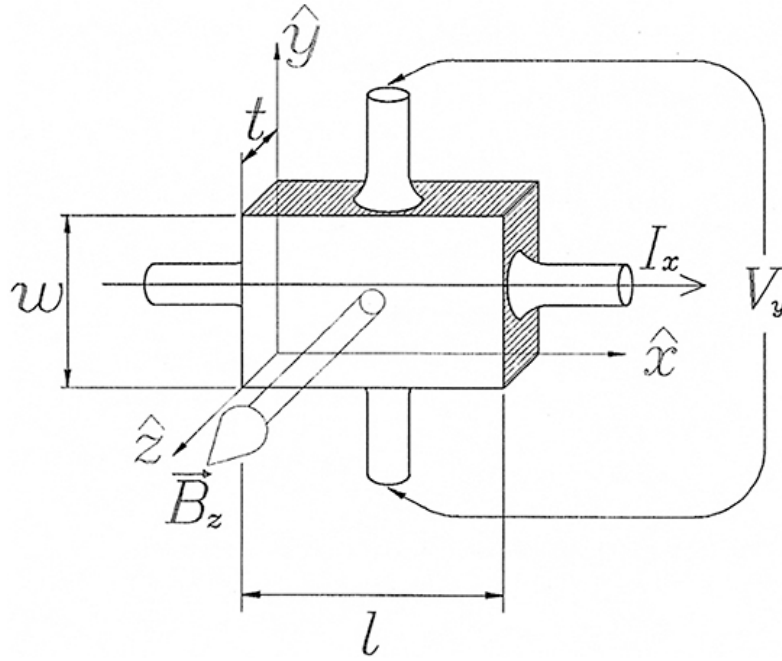


Figure 3.2. Typical Hall experiment setup showing direction of the current, I_x and the magnetic field B_z . V_y denotes the Hall voltage.

At very low temperature, or at very high values of the magnetic field (or at both), the resistivity of the sample assumes quantized values of the form, $\rho_{xy} = \frac{h}{ne^2}$. Initially n was found to be an integer with extraordinary precision (one part in $\sim 10^8$). This is shown in figure 3.3. The quantization of the Hall resistivity yields the name 'quantum' (or quantized) Hall effect, which we refer to as QHE throughout the chapter.

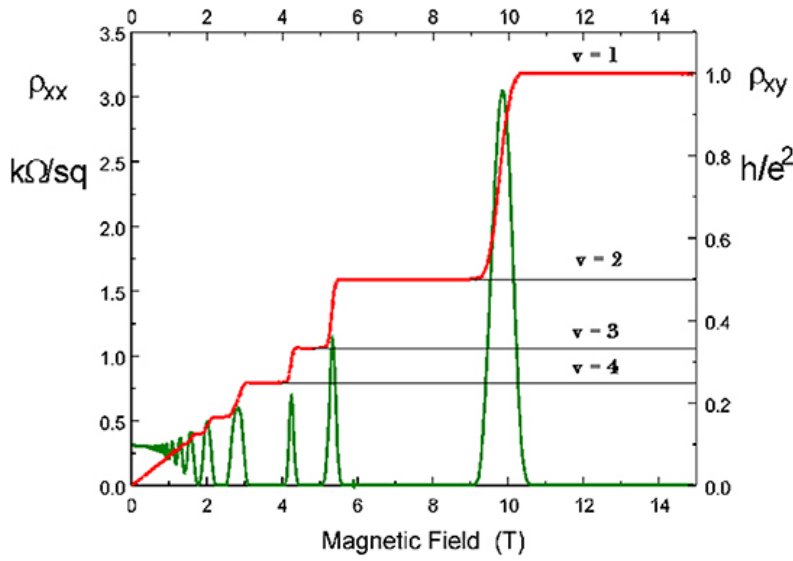


Figure 3.3. Schematic plot of integer quantum Hall effect (IQHE) as a function of the applied magnetic field. The plot in red denotes the Hall resistivity and the one in green shows the longitudinal resistivity (or the magnetoresistivity).

Klaus von Klitzing and his co-workers [13, 14], while measuring the electrical transport properties of planar systems formed at the interface of two different semiconducting samples in the strong magnetic field facility at Grenoble, France, noted that the Hall resistivity is quantized in units of h/e^2 as a function of the external magnetic field. The flatness of the plateaus occurring at integer or fractional values of h/e^2 has an unprecedented precision and is independent of the geometry of the sample (as long as it is 2D), density of the charge carriers and its purity. The accuracy of the quantization aids in fixing the unit of resistance, namely, $h/e^2 = 25.813 \text{ K}\Omega$, also called as the Klitzing constant. Thus, among other significant properties of QHE that we shall be discussing in due course, an experiment performed at a macroscopic scale that can be used for metrology or yields the values for the fundamental constants used in quantum physics is truly amazing and hence calls for intense scrutiny. The effect occurs when the density of the carriers, n are such that they are encoded in the integers (that come as proportionality constants to the Hall resistivity in terms of h/e^2) as if the charges locked their separation at some particular values. The phenomenon remains resilient to changing the carrier density by a small amount, however, changing it by a large amount does destroy the effect.

The Hall resistivity (in red) in figure 3.3 becomes constant for certain ranges of the external magnetic fields, which are called plateaus. Further the longitudinal resistivity (in green) in the same plot vanishes everywhere. Although it shows peaks wherever there is a jump in the Hall resistivity from one plateau to another. Later on it was found that ν is not only restricted to integer values, but also takes values which are rational fractions, such as $\nu = \frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{3}{7}, \frac{4}{9}, \frac{5}{9}$, etc. There are about 100 fractions (including the improper ones) that have been noted in experiments so far. The corresponding plot appears in figure 3.4.

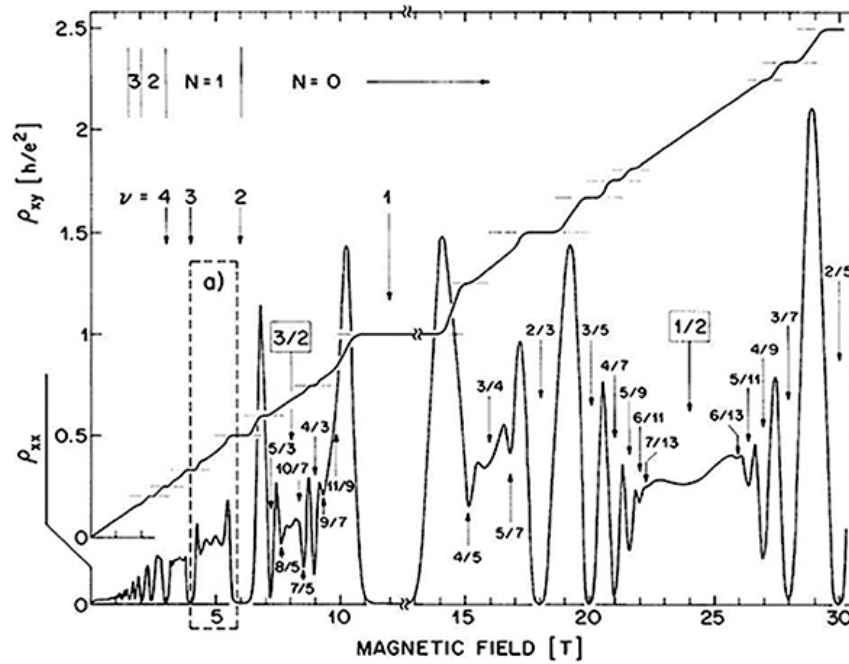


Figure 3.4. The plot shows fractional quantum Hall effect (FQHE). The plateaus are shown at fractional values in units of h/e^2 . Reprinted with permission from [15], copyright (1982) by the American Physical Society.

3.2.1 General perspectives

The charge carriers being confined in 2D wells has a longer history. Since 1966 it has been known that the electrons accumulated at the surface of a silicon single crystal induced by a positive gate voltage form a 2D electron gas (2DEG). The energy of the electrons corresponding to a motion perpendicular to the surface is quantized (box quantization) and on top of it, the free motion of the electrons in 2D becomes quantized when a strong magnetic field is applied perpendicular to the plane (Landau quantization). Thus, QHE has both the quantization phenomena in-built into it.

An important recent development in the study of semiconductors is the achievement of structures in which the electrons are restricted to move essentially in 2D. This immediately says that the carriers are prohibited from moving along the direction transverse to the plane. Hence the motion is quantized. Such 2D behaviour of the carriers can be found in metal-oxide semiconductor (MOS) structures, quantum wells and superlattices. An excellent prototype is the metal-insulator-semiconductor (MIS) layered structure of which the insulator is usually an oxide, such as Al_2O_3 (thereby making it a MOS structure). In figure 3.5 we show a typical MOS device where the substrate is a doped p -type silicon (Si) which is grounded and is called a base (shown by B in figure 3.5). On the top, there is a metallic layer (shown by the hatched regime) followed by an insulating layer formed by SiO_2 . The metallic layer is called the gate, denoted by G which is biased by a voltage V_G . The source (grounded) and the drains are denoted by S and D, respectively, which in figure 3.5 consist of n -type materials. The gate voltage causes the carriers beneath the gate electrode to drift between the source and the drain. The layer of charge carriers below the oxide layer forms the 2DEG which is central to our discussion. The energy dispersion in this case reads,

$$E_n(k_x, k_y) = \frac{\hbar^2 k_x^2}{2m_{xx}^*} + \frac{\hbar^2 k_y^2}{2m_{yy}^*} \quad (3.2)$$

where m_{xx}^* and m_{yy}^* are the components of the effective mass tensor defined by the inverse of the curvature of the band structure, namely,

$$m_{\alpha\beta}^* = \hbar^2 \left(\frac{\partial^2 E(k)}{\partial k_\alpha \partial k_\beta} \right)^{-1}. \quad (3.3)$$

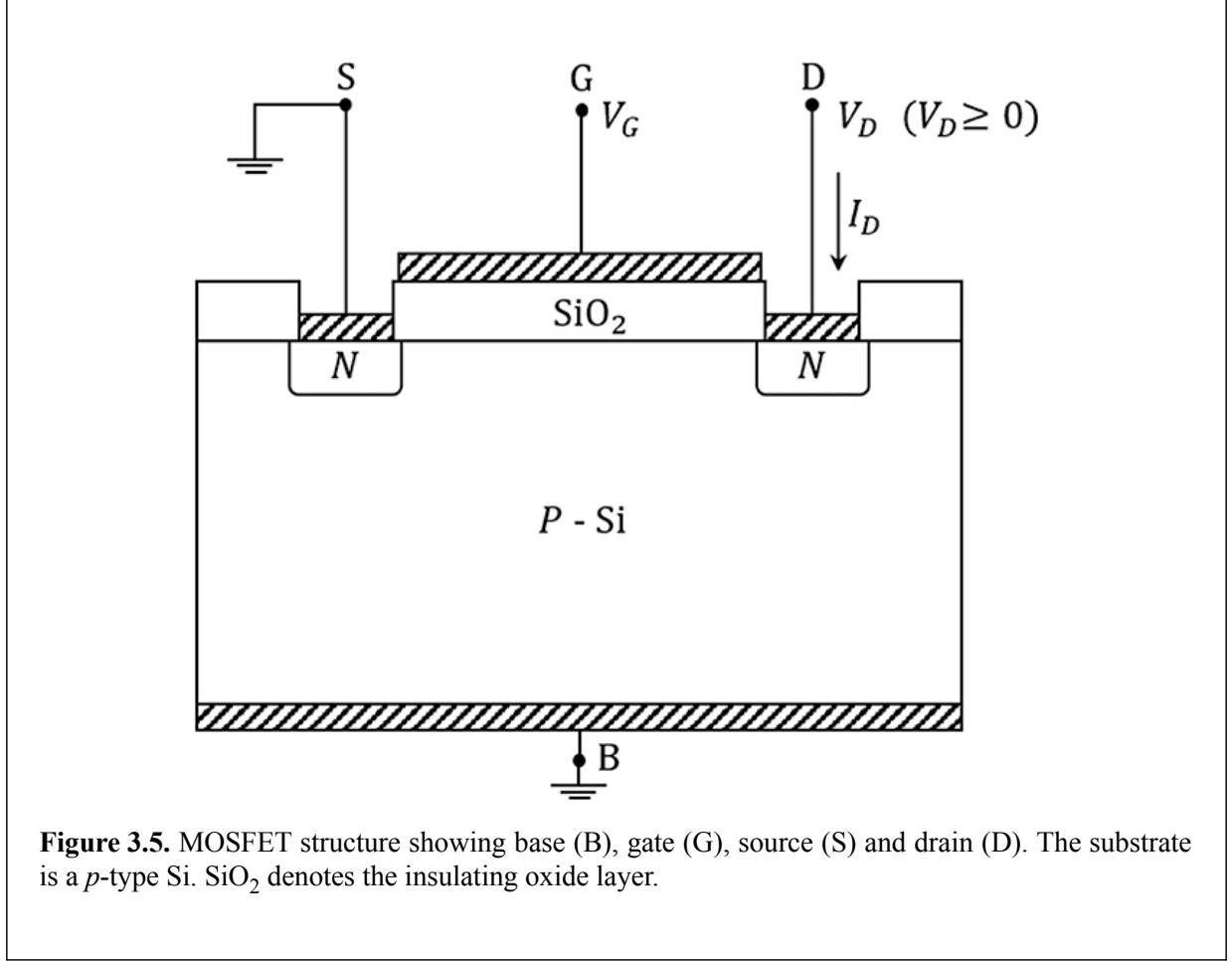


Figure 3.5. MOSFET structure showing base (B), gate (G), source (S) and drain (D). The substrate is a *p*-type Si. SiO₂ denotes the insulating oxide layer.

The physical properties of all systems are governed by their density of states (DOS) which plays a crucial role in deciding on the dependence of the temperature, density of carriers, etc. In 2D systems with a parabolic dispersion as elaborated above in equation (3.2), the DOS is a constant and assumes a form¹,

$$g(E) = g_{2D}(E) = \frac{m^*}{\pi \hbar^2}. \quad (3.4)$$

The energy independent DOS is very special to 2D and is in sharp contrast to three-dimensional (3D) where it goes as $E^{1/2}$, and in one-dimensional (1D) where it goes as $E^{-1/2}$. In a general sense, and not restricted to the discussion on Hall effect, the DOS enters while calculating the average quantities, such as

the average energy or the average number of particles. For example, the average of a physical observable, O of a fermionic system is computed using,

$$\langle O \rangle = \int_0^\mu O f(E) g(E) dE \quad (3.5)$$

where $f(E)$ is the Fermi distribution function given by,

$$f(E) = \frac{1}{e^{\beta(E-\mu)} + 1}$$

with $\beta = \frac{1}{k_B T}$ and μ denotes the chemical potential. In general this integral is quite challenging to compute analytically because of the Fermi distribution function ($f(E)$) present in the integrand.

Meanwhile, there is a wonderful simplification where $f(E)$ assumes a value unity at all temperatures for which the experiments are performed. Only at temperatures close to the Fermi temperature, T_F defined via, $\epsilon_F = k_B T_F$ (ϵ_F being the Fermi energy), $f(E)$ starts to deviate from unity, and its exact form needs to be incorporated in the integral. However, T_F is usually of the order of tens of thousands of Kelvin for typical metals (such as, Cu, Al, etc) which is too high for them to appear in experimental situations. Moreover, the DOS only depends on energy and is independent of the temperature to a very good approximation. Thus computation of equation (3.5) becomes trivial as the integrand becomes independent of temperature or weakly dependent on temperature [16, 17].

In the following, we mention that there is something interesting about transport properties of 2D systems. In the linear response regime Ohm's law is valid, and says that

$$V_\alpha = R_{\alpha\beta} I_\beta$$

where $R_{\alpha\beta}$ denotes the resistivity tensor, and α, β denote spatial variables, x, y , etc. One can equivalently invert this equation to write, $I_\alpha = G_{\alpha\beta} V_\beta$, where $G_{\alpha\beta}$ represents the conductivity tensor with $G = R^{-1}$. Equivalent relations in terms of the components of the electric field (E) and current density (j) read as,

$$E_\alpha = \rho_{\alpha\beta} j_\beta \quad \text{and} \quad j_\alpha = \sigma_{\alpha\beta} E_\beta \quad (3.6)$$

where ρ and σ denote the resistivity and the conductivity tensors, respectively.

An interesting (and useful too) artefact of 2D physics is an accidental similarity that exists in decoding some of the key features of the transport properties. For example, the resistivity, ρ (or the conductivity, σ) is a quantity that is independent of the system geometry, and hence is useful for a theoretical analysis, whereas in experiments, one measures the resistance of a sample, R (or the conductance, G). For a sample in the shape of a hypercube of sides, L , the resistance and the resistivity are related by,

$$R = \rho L^{2-d} \quad (3.7)$$

where d denotes the dimensionality. Only for $d = 2$ the resistance is a scale invariant quantity. This puts the experimentalists and the theorists on the same page, as the geometry of the sample does not enter explicitly in the analysis of its transport properties.

As we dig more into the details of transport properties of 2DEG in the presence of a magnetic field, further useful information emerges. The off-diagonal elements of both the conductivity and the resistivity tensors are antisymmetric with regard to the direction of the applied field, $\mathbf{B}$. Consider a planar sample with dimensions $L_x \times L_y$. The conductivity tensor is of the form,

$$\sigma = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{pmatrix}. \quad (3.8)$$

Let us try to understand the nature of the tensor, σ in the presence of a magnetic field. A conductor in an external magnetic field obeys,

$$j_\alpha = \sigma_{\alpha\beta} E_\beta. \quad (3.9)$$

Onsager's reciprocity principle does not hold in the presence of a magnetic field, $\mathbf{B}$, which implies [18],

$$\sigma_{\alpha\beta}(\mathbf{B}) \neq \sigma_{\beta\alpha}(\mathbf{B}). \quad (3.10)$$

Instead one has, $\sigma_{\alpha\beta}(\mathbf{B}) = \sigma_{\beta\alpha}(-\mathbf{B})$ to make sure that the time reversal holds only if $\mathbf{B}$ changes sign. Let us write the conductivity tensor as a sum of a symmetric and an antisymmetric tensor (note that this is always possible for a rank two tensor). Thus,

$$\sigma_{\alpha\beta} = S_{\alpha\beta} + A_{\alpha\beta} \quad (3.11)$$

where $\mathbf{S}$ and $\mathbf{A}$ are the symmetric and the antisymmetric tensors which obey the following relations,

$$\begin{aligned} S_{\alpha\beta}(\mathbf{B}) &= S_{\beta\alpha}(-\mathbf{B}) = S_{\alpha\beta}(-\mathbf{B}) \\ A_{\alpha\beta}(\mathbf{B}) &= A_{\beta\alpha}(-\mathbf{B}) = -A_{\alpha\beta}(-\mathbf{B}) \end{aligned} \quad (3.12)$$

such that the components of $S_{\alpha\beta}$ are even functions of $\mathbf{B}$, while those of $A_{\alpha\beta}$ are odd functions of $\mathbf{B}$. Putting in equation (3.9),

$$j_\alpha = S_{\alpha\beta} E_\beta + A_{\alpha\beta} E_\beta. \quad (3.13)$$

But owing to the antisymmetry,

$$A_{\alpha\beta} = \epsilon_{\gamma\alpha\beta} A_\gamma = -\epsilon_{\alpha\gamma\beta} A_\gamma. \quad (3.14)$$

Putting it in equation (3.13),

$$\begin{aligned} j_\alpha &= S_{\alpha\beta} E_\beta - \epsilon_{\alpha\beta\gamma} A_\beta E_\gamma \\ &= S_{\alpha\beta} E_\beta - (\mathbf{A} \times \mathbf{E})_\alpha \\ &= S_{\alpha\beta} E_\beta + (\mathbf{E} \times \mathbf{A})_\alpha. \end{aligned} \quad (3.15)$$

Assuming that we can expand $\sigma(\mathbf{B})$ in powers of $\mathbf{B}^2$, such that the antisymmetric part contains odd powers of $\mathbf{B}$, then we can write,

$$A_\alpha = \eta_{\alpha\beta} B_\beta \quad (3.16)$$

and $S_{\alpha\beta}(\mathbf{B})$ consists of even powers of $\mathbf{B}$,

$$S_\alpha = (\sigma_0)_{\alpha\beta} + \zeta_{\alpha\beta\gamma\delta} B_\gamma B_\delta. \quad (3.17)$$

The first term is the zero field conductivity tensor. Thus, putting things together up to terms linear in $\mathbf{B}$,

$$j_\alpha = S_{\alpha\beta} E_\beta + (\mathbf{E} \times \mathbf{A})_\alpha.$$

The second term denotes the Hall effect which is linear in $\mathbf{B}$. This implies that the Hall current is perpendicular to the electric field, $\mathbf{E}$ and is proportional to $\mathbf{E}$ and $\mathbf{B}$. Thus an antisymmetric tensor is relevant to the study of the Hall effect, which is why the conductivity and the resistivity tensors are antisymmetric. This is an important result which deviates from the corresponding scenario that arises in the absence of an external magnetic field.

3.2.2 Translationally invariant system: classical limit of QHE

It is quite an irony that the extreme universal signature of the transport properties of a 2DEG are characterized by the flatness of the plateaus, that not only occurs, but survives even in the presence of disorder, impurity and imperfection. In the absence of the magnetic field, Anderson localization would have governed the transport signatures of non-interacting electrons which says that in any dimension less than three, all eigenstates of a system are exponentially localized even for an infinitesimal disorder strength. Only in three dimensions is there a critical disorder at which a metal-insulation transition occurs. However, the scenario is strongly altered by the presence of the magnetic field, which yields, as we shall shortly see, a series of unique phase transitions from a perfect conductor to a perfect insulator. No other system demonstrates re-occurrence of the same phases over and over again as the magnetic field is gradually ramped up.

To begin with, we shall consider the case which is free from disorder, or equivalently a translationally (Lorentz) invariant system that possesses no preferred frame of reference. Thus we can think of a reference frame that is moving with a velocity $-\mathbf{v}$ with respect to the lab frame, where the current density is given by, $\mathbf{j} = -ne\mathbf{v}$ (n : areal electron density, $-e$: electronic charge). In this frame, the electric and the magnetic fields are given by³,

$$\mathbf{E} = -\mathbf{v} \times \mathbf{B} ; \quad \text{and} \quad \mathbf{B} = B\hat{z}. \quad (3.19)$$

The above transformation ensures that an electric field must exist to balance the Lorentz force $-\mathbf{ev} \times \mathbf{B}$ in order to conduct without deflection. For the electric field, this yields,

$$\mathbf{E} = \frac{1}{ne} \mathbf{J} \times \mathbf{B}. \quad (3.20)$$

This is equivalent to the tensor equation,

$$E^\mu = \rho_{\mu\nu} j^\nu \quad (3.21)$$

with the resistivity tensor given by,

$$\rho_{\mu\nu} = \frac{B}{ne} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (3.22)$$

Inverting the tensor equation, one can obtain,

$$j^\mu = \sigma_{\mu\nu} E^\nu \quad (3.23)$$

where the conductivity tensor, $\sigma_{\mu\nu}$ is defined via,

$$\sigma_{\mu\nu} = \frac{ne}{B} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad (3.24)$$

There is an interesting paradox which states that, $\sigma_{xx} = \rho_{xx} = 0$ (see above) which are of course contradictory. However, we reserve this rather interesting topic for a discussion immediately afterwards. Here we wish to point out that we get the results for a classical Hall effect, that is, $\sigma_{xy} = \frac{ne}{B}$ (or $\rho_{xy} = \frac{B}{ne}$). It is important to realize that the result is an artefact of Lorentz invariance, where the characteristics of the sample or the 2DEG enter only through the carrier density, n for a translationally invariant system. Thus, in the absence of defect, disorder and impurity, the Hall effect concerns with the carrier density of the sample and nothing else. The Hall resistivity depends linearly on the magnetic field. The QHE which is much more versatile (than merely depending on the density) where disorder, that jeopardizes the translational invariance plays an indispensable part in causing plateaus in the conductivity (or the resistivity) to occur and survive. We show a schematic plot in figure 3.6 to emphasize the difference between the classical and the QHE. Both the Hall and the longitudinal resistivities are significantly different in these two cases.

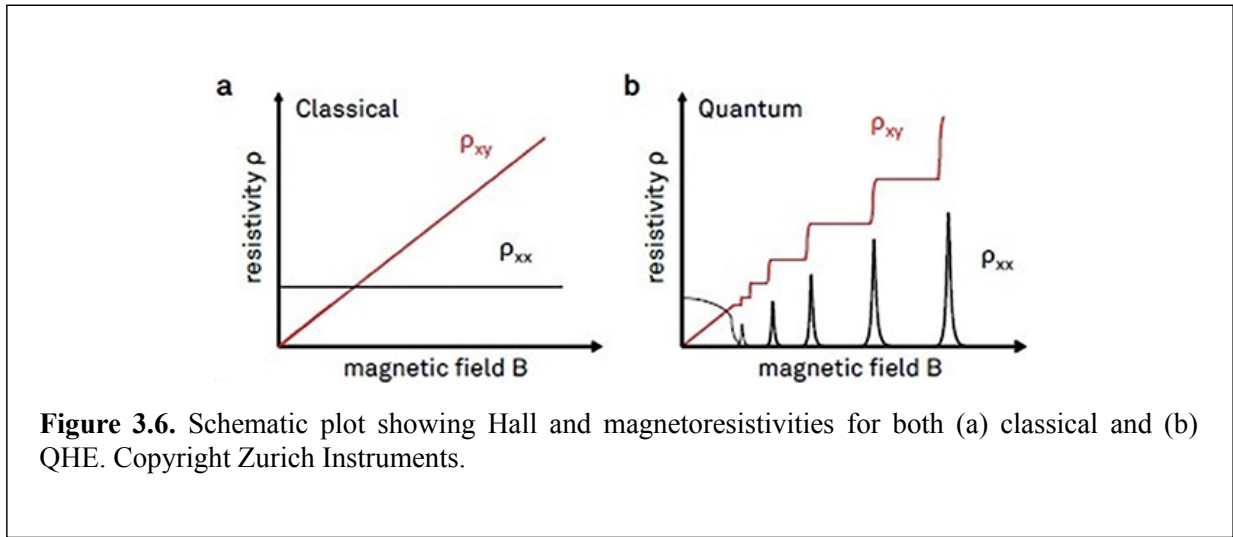


Figure 3.6. Schematic plot showing Hall and magnetoresistivities for both (a) classical and (b) QHE. Copyright Zurich Instruments.

Writing the equation of motion (EOM) for a charge particle of mass m , moving with a velocity, $\mathbf{v}$ ($= \frac{\mathbf{p}}{m}$) in presence of a longitudinal electric field, $\mathbf{E}$ and a perpendicular magnetic field, $\mathbf{B}$,

$$\frac{d\mathbf{p}}{dt} = -e\mathbf{E} - e\frac{\mathbf{p}}{m} \times \mathbf{B} - \frac{\mathbf{p}}{\tau}. \quad (3.25)$$

The last term is the resistive force arising from electron-impurity scattering with τ being the relaxation time. Using the current density, $\mathbf{j} = -ne\mathbf{p}/m$ and the cyclotron frequency, $\omega_B = eB/m$, at the steady state ($\frac{d\mathbf{p}}{dt} = 0$),

$$\frac{ne^2\tau}{m}\mathbf{E} + \omega_B\tau\mathbf{j} \times \hat{\mathbf{z}} + \mathbf{j} = 0. \quad (3.26)$$

Assuming the motion of carriers along the x -direction, that is, $\mathbf{j} = j\hat{\mathbf{x}}$, casting it in the form, $\mathbf{j} = \sigma\mathbf{E}$, σ assumes the form,

$$\sigma = \frac{ne^2\tau/m}{1 + \omega_B^2\tau^2} \begin{pmatrix} 1 & -\omega_B\tau \\ \omega_B\tau & 1 \end{pmatrix}. \quad (3.27)$$

This yields the Drude conductivity which can be written as,

$$\sigma_{xx} = \frac{\sigma_0}{1 + \omega_B^2 \tau^2} \quad (3.28)$$

where $\sigma_0 = ne^2\tau/m$. In the absence of any scattering by the impurities, the relaxation time, τ is infinitely large, which yields $\sigma_{xx} \rightarrow 0$. This induces Ohm's law to assume the form (now writing in terms of the resistivity),

$$\begin{aligned} \mathbf{E} = \begin{pmatrix} E_x \\ E_y \end{pmatrix} &= \begin{pmatrix} 0 & \rho_{xy} \\ -\rho_{xy} & 0 \end{pmatrix} \begin{pmatrix} j_x \\ j_y \end{pmatrix} \\ &= \begin{pmatrix} \rho_{xy} j_y \\ -\rho_{xy} j_x \end{pmatrix}. \end{aligned} \quad (3.29)$$

Hence the electric field $\mathbf{E}$ is perpendicular to the current density, $\mathbf{j}$, which says that, $\mathbf{j} \cdot \mathbf{E} = 0$. The physical significance of $\mathbf{j} \cdot \mathbf{E}$ is the work done that accelerates the charges, which being zero in this case implies that a steady current flows in the sample without requiring any work, and hence causes no dissipation. Thus $\sigma_{xx} = 0$ implies that no current flows in the longitudinal direction, which is actually a signature of a perfectly insulating state. Since the components of the resistivity and the conductivity tensors are related by,

$$\sigma_{xx} = \frac{\rho_{xx}}{\rho_{xx}^2 + \rho_{xy}^2} \quad ; \quad \sigma_{xy} = \frac{-\rho_{xy}}{\rho_{xx}^2 + \rho_{xy}^2}. \quad (3.30)$$

Let us examine the possible scenarios:

- (i) If $\rho_{xy} = 0$, one gets $\sigma_{xx} = \frac{1}{\rho_{xx}}$ and $\sigma_{xy} = 0$ which is a familiar scenario.
- (ii) If $\rho_{xy} \neq 0$, σ_{xx} and σ_{xy} both exist.
- (iii) Now consider $\rho_{xx} = 0$, $\sigma_{xx} = 0$, if $\rho_{xy} \neq 0$. This is truly interesting, since if $\rho_{xx} = 0$, it implies a perfect conductor, and at the same time $\sigma_{xx} = 0$ implies a perfect insulator. This is surprising, but truly occurs in the presence of an external magnetic field. This is reflected in the plots presented in figures 3.3 and 3.4. We shall return for a more thorough discussion later.

3.2.3 Charge particles in a magnetic field: Landau levels

Let us examine the fate of the electrons confined in a 2D plane in the presence of a magnetic field. Consider non-interacting spinless electrons in an external field, $\mathbf{B}$ and write down the Schrödinger equation to solve for the eigenvalues and eigenfunctions. The canonical momentum can now be written as, $\mathbf{p} \rightarrow \mathbf{p} - q\mathbf{A} = \mathbf{p} + e\mathbf{A}$ [19] where $q = -e$ is the electronic charge and $\mathbf{A}$ is the vector potential corresponding to the field, $\mathbf{B}$. $\mathbf{B}$ and $\mathbf{A}$ are related via, $\nabla \times \mathbf{A} = \mathbf{B}$. The time-independent Schrödinger equation becomes,

$$\frac{1}{2m}(\mathbf{p} + e\mathbf{A})^2 \psi(\mathbf{r}) = E\psi(\mathbf{r}). \quad (3.31)$$

In order to solve this, we need to fix a gauge or a choice of the vector potential. Note that the choice of the gauge will not alter the solution of the equation, which in other words can be stated as the Schrödinger equation being gauge invariant. However, a particular choice is essential for us to go ahead.

Corresponding to a magnetic field, $\mathbf{B}$ in the z -direction, such that $\mathbf{B} = B\hat{z}$ (just as the case of a 2DEG subjected to a perpendicular magnetic field), the vector potential can be chosen as,

$$A_x = -By, \quad A_y = A_z = 0.$$

This is known as the Landau gauge. It also allows us to assume $A_y = Bx$ and $A_x = A_z = 0$ ⁴. In the Landau gauge, the Schrödinger equation becomes,

$$\left[\frac{1}{2m} (p_x - eBy)^2 + \frac{p_y^2}{2m} + \frac{p_z^2}{2m} \right] \psi(\mathbf{r}) = \epsilon \psi(\mathbf{r}). \quad (3.33)$$

Clearly, in the z -direction, the particle behaves like a free particle with energy, $\epsilon_z = \frac{p_z^2}{2m}$ with the eigenfunction the same as that of particle in a box in the z -direction. Thus in the x - y plane, the above equation becomes,

$$\left[\frac{1}{2m} (p_x - eBy)^2 + \frac{p_y^2}{2m} \right] g(x, y) = \epsilon g(x, y) \\ \mathcal{H}(x, y) g(x, y) = \epsilon g(x, y) \quad (3.34)$$

where for the 2D case, $\psi(\mathbf{r})$ becomes $g(x, y)$ and $E = \epsilon + \epsilon_z$. It is easy to see that p_x commutes with $\mathcal{H}(x, y)$, that is,

$$[\mathcal{H}(x, y), p_x] = 0. \quad (3.35)$$

Hence p_x is a constant of motion. Thus for a p_x given by, $p_x = \frac{2\pi\hbar}{L_x} n_x$, one can write equation (3.34) as,

$$\left[\frac{p_y^2}{2m} + \frac{1}{2} m \left(\frac{eB}{m} \right)^2 (y - y_0)^2 \right] f(y) = \epsilon f(y) \quad (3.36)$$

where $y_0 = \frac{p_x}{eB} = k_x l_B^2 = k l_B^2$ (say), and $f(y)$ is only a function of y . y_0 has the dimension of length, and hence l_B is denoted as the magnetic length which is an important quantity in all of subsequent discussions. Moreover, $f(y)$ is the eigenfunction corresponding to the 1D Hamiltonian written above.

Interestingly, the left-hand side of equation (3.36) denotes the Hamiltonian for a simple harmonic oscillator (SHO) which oscillates in the y -direction about a mean position y_0 with a frequency, $\omega_c = \frac{eB}{m}$. ω_c is known as the cyclotron frequency. Taking results from the SHO problem in quantum mechanics, the energy eigenvalues can be found as,

$$\epsilon_n = \left(n + \frac{1}{2} \right) \hbar \omega_c = \left(n + \frac{1}{2} \right) \hbar \frac{eB}{m}. \quad (3.37)$$

The energy levels in equation (3.37) are referred to as the Landau levels. The levels are schematically shown in figure 3.7. They are equidistant for a given value of the magnetic field, however their separation increases with the increase in B . Each of these levels has a large degeneracy (see below).

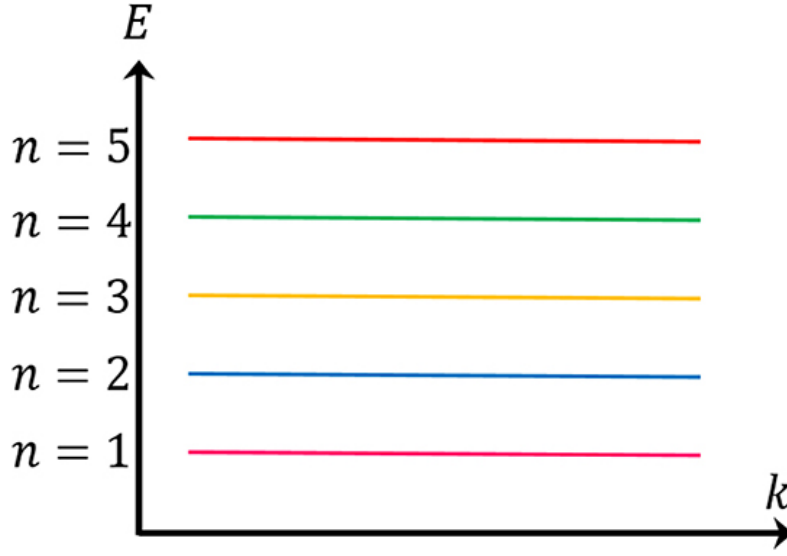


Figure 3.7. Schematic plot showing the Landau levels. Each of these levels has a large degeneracy.

Further, the eigenfunction, $g(x, y)$ in equation (3.34) corresponding to an oscillatory motion (as an SHO) in the y -direction and a free motion (like a particle in the absence of any potential) in x -direction assumes the form,

$$g(x, y) = \frac{1}{L_x} e^{ik_x x} A_n e^{-\frac{eB(y-y_0)^2}{\hbar}} H_n \left(\frac{eB(y-y_0)}{\hbar} \right) \quad (3.38)$$

where $H_n(\xi)$ with $\xi = \frac{eB(y-y_0)}{\hbar}$ denote the Hermite polynomials that are familiar in the context of SHO and A_n denote the normalization constants. Thus the trajectory of the particle is similar to that of a simple harmonic oscillator centred about a certain value of y (instead of the origin), namely, y_0 and freely propagating along the x -direction. y_0 is controlled by the strength of the magnetic field B , and is inversely proportional to it.

3.2.4 Degeneracy of the Landau levels

The Landau levels given by equation (3.37) are hugely degenerate. Since p_x is a constant of motion, the energy is independent of p_x . Thus all possible values of the quantum numbers corresponding to the motion in the x -direction, namely, n_x that are defined by, $k_x = \frac{2\pi}{L_x} n_x$, (L_x denotes the length of the sample in the x -direction and $n_x = 0, 1, 2, \dots$) will make the levels degenerate. The degree of degeneracy is only limited by the length of the sample in the y -direction, namely, L_y . Since the magnetic length, $y_0 = kl_B^2 = k \left(\frac{\hbar}{eB} \right)$ about which the simple harmonic motion occurs should not exceed the length L_y of the sample, the maximum degeneracy of the Landau levels can be found by substituting $y_0 = L_y = \left(\frac{\hbar}{eBL_x} \right) n_x$, and hence using the maximum possible value of n_x , that is,

$$(n_x)_{\max} = g = \frac{eBL_x L_y}{h} = \frac{eBA}{h}$$

where $\mathcal{A} = L_x L_y$ is the area of the sample in the x - y plane. This yields the degeneracy, g to be identical to the flux, $\phi (= B\mathcal{A})$ threading the planar sample, via, $g = \Phi/\Phi_0$ where $\Phi_0 = h/e^5$. (3.39)

A few comments are in order:

- (i) The degeneracy, g is independent of the effective mass of the carriers, and hence independent of the material.
- (ii) The degeneracy is proportional to the area of the sample and the value of the magnetic field. Thus the degeneracy can be controlled by the applied magnetic field.

To remind ourselves, we have solved for the properties of a single electron confined in a plane in the presence of a perpendicular magnetic field. The energies are called the Landau levels, and these levels are highly degenerate. The scenario is a prototype of what happens in a Hall effect experiment where the external magnetic field is varied and the resistivities (both the Hall and the longitudinal) are measured. The Hall resistivity shows plateaus in multiples of $h/n\epsilon^2$ whenever the filling fraction, ν of the Landau levels (defined below) is close to an integer n . Consider n_0 to be the density of charge carriers of the sample, then ν is defined using, $\nu = \frac{n_0}{g/\mathcal{A}} = \frac{n_0 h}{eB}$. ν denotes the filling fraction which is also defined as,

$$\nu = \frac{\text{number of electrons}}{\text{flux quantum}}$$

As and when ν assumes a value near an integer, n (or a rational fraction as in the case of FQHE) as the magnetic field is tuned, one observes a plateau in the Hall resistivity, ρ_{xy} . Something else happens at the same time that is equally interesting. The longitudinal resistivity, ρ_{xx} drops to zero whenever ρ_{xy} acquires a plateau. The vanishing of ρ_{xx} makes the system dissipationless. However, the diagonal conductivity (σ_{xx}) also vanishes which makes the system insulating. The paradox is explained in detail elsewhere.

3.2.5 Conductivity of the Landau levels: role of the edge modes

Here we discuss some of the key properties of the Landau levels. Let us now calculate the current carried by the Landau levels. The expression for the current can be found using $\langle \mathbf{J} \rangle = -e\langle \mathbf{v} \rangle$ where the expectation value has to be computed within the Landau states.

$$\langle \mathbf{J} \rangle = -e\langle \psi_k | \mathbf{v} | \psi_k \rangle = -\frac{e}{m}\langle \psi_k | \mathbf{p} + e\mathbf{A} | \psi_k \rangle. \quad (3.40)$$

The longitudinal current in the x -direction carried by the Landau levels is obtained via,

$$\langle J_x \rangle = -\frac{e}{ml_B\sqrt{\pi}} \int dy e^{-\frac{1}{l_B^2}(y-y_0)^2} (\hbar k - eBy) = 0. \quad (3.41)$$

The integrand has an even function (the first one) and an odd function (the second one). Thus the integral vanishes. We can get the average velocity, $\langle v \rangle = \frac{1}{\hbar} \frac{\partial \epsilon_k}{\partial k} = 0$ as ϵ does not depend upon k . Thus the Landau wavefunctions carry no current by themselves. They only carry current in the presence of an electric field in the x -direction as we show below.

3.2.6 Spin and the electric field

So far we have been talking about spinless fermions. It is in general a worthwhile exercise to include the spin of the electrons and explore if there is any significant development to the quantization phenomena discussed above. The spin degrees of freedom placed in an external magnetic field introduce a Zeeman

energy scale owing to a coupling between the spin and the magnetic field. The Zeeman term is written as $\Delta_Z = g\mu_B B$, where $\mu_B = e\hbar/2m$ is the Bohr magneton and $g = 2$. Now the splitting between the Landau levels originating from the orbital effect ($\mathbf{p} \rightarrow \mathbf{p} - q\mathbf{A}$) is $\Delta = \hbar\omega_B = \frac{e\hbar B}{m} = \Delta_B$ (say). Δ_B is the so called cyclotron energy. But for electrons, this precisely coincides with the Zeeman splitting $\Delta = g\mu_B B$ between the $\uparrow$ and the $\downarrow$ -spins. Thus it looks that the spin- $\uparrow$ particles in a Landau level with index, n have exactly the same energy as the spin- $\downarrow$ particles in the next higher Landau level index, that is, $n + 1$. However, in real materials this does not occur. For example in GaAs, the Zeeman energy is typically about 70 times smaller than the cyclotron energy.

Now consider an external electric field, E applied in the x -direction. This creates an electric potential of the form, $\phi = -Ex$. Thus the resulting Hamiltonian including external electric and magnetic fields can be written as,

$$\mathcal{H} = \frac{1}{2m} \left[p_x^2 + (p_y + eBx)^2 \right] + eEx. \quad (3.42)$$

It is important to note that we have considered a different choice of the gauge here, namely, $A_y = Bx$. Instead of a rigorous derivation, we can complete square in the expression for energy of the particles,

$$E_{n,k} = \hbar\omega_B \left(n + \frac{1}{2} \right) - eE \left(kl_B^2 + \frac{eE}{m\omega_B^2} \right) + \frac{1}{2} m \frac{E^2}{B^2}. \quad (3.43)$$

This is interesting because the degeneracy of the Landau level has now been lifted. The energy in each level now depends linearly on k as shown in figure 3.8, which was earlier independent of k . The eigensolution is simply that of a harmonic oscillator shifted from the origin and displaced along the x -axis by an amount mE/eB^2 and is written as,

$$\psi(x, y) = \psi_{n,k} \left(x + \frac{mE}{eB^2}, y \right). \quad (3.44)$$

Among the other properties, the group velocity is given by,

$$v_y = \frac{1}{\hbar} \frac{\partial E_{n,k}}{\partial k} = \frac{e}{\hbar} El_B^2. \quad (3.45)$$

Putting $l_B = \sqrt{\frac{\hbar}{eB}}$ (as said earlier, l_B is an important length scale of the problem which we shall see throughout the discussion),

$$v_y = \left(\frac{eE}{\hbar} \right) \cdot \left(\frac{\hbar}{eB} \right) = \frac{E}{B}. \quad (3.46)$$

Thus the energy has three terms in equation (3.43),

- (i) the first one is that of a harmonic oscillator,
- (ii) the second one is the potential energy, of a wave packet localized at $x = \left(-kl_B^2 - \frac{mE}{e\omega_B^2} \right)$,
- (iii) and finally the last one denotes the kinetic energy of the particle, namely, $\frac{1}{2}mv_y^2$.

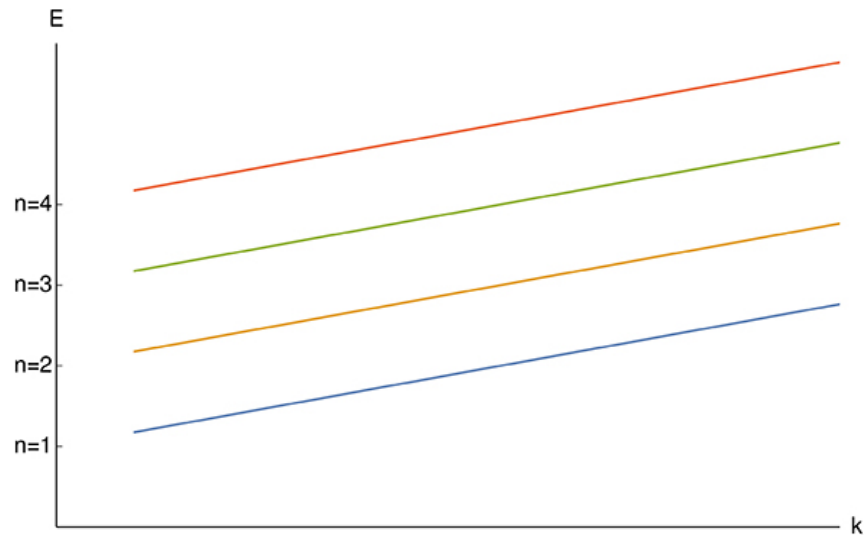


Figure 3.8. Schematic plot showing the Landau levels ($n = 1, 2, 3, 4$) in the presence of an electric field. The levels are tilted because of the electric field.

3.2.7 Laughlin's argument: Corbino ring

Laughlin intuitively considered the phenomenon of QHE as a quantum pump. Consider a ring where the vacant region admits a magnetic field and hence a flux ϕ . For the argument to be valid, the geometry of the ring is important. Here in addition to the background magnetic field $\mathbf{B}$ that threads the sample, we can thread an additional flux Φ through the center of the ring (see figure 3.9). This Φ can affect the quantum state of the electrons. In addition, the temperature is low such that the thermal effects can be neglected.

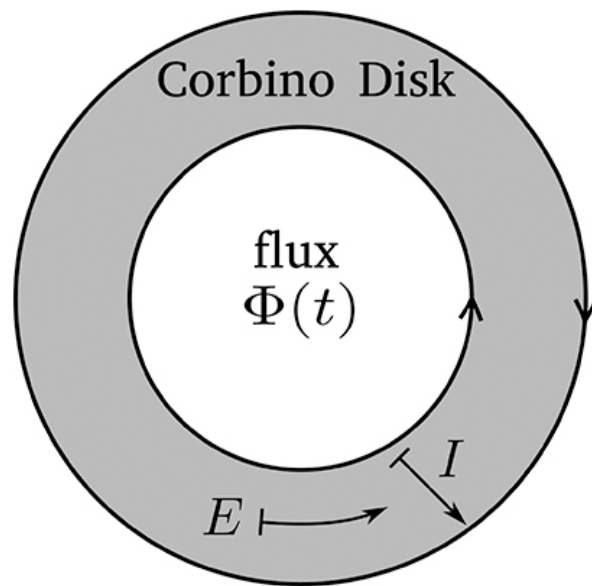


Figure 3.9. Schematic plot showing Corbino ring.

Let us first see what this flux Φ has got to do with the Hall conductivity. Suppose we slowly increase Φ from 0 to $\Phi_0 (= \frac{h}{e})$, that is, within a time $t_0 \gg \frac{1}{\omega_B}$. This induces an emf around the annular region $\varepsilon = \frac{\partial \Phi}{\partial t} = \frac{-\Phi_0}{t_0}$. The purpose of this emf is to transport ‘ n ’ electrons from the inner circumference to the outer circumference. This would result in a current in the radial direction, $I_r = -ne/t_0$. Thus the Hall resistivity is,

$$\rho_{xy} = \frac{\varepsilon}{I_r} = -\frac{\Phi_0}{t_0} \cdot \frac{t_0}{(-ne)} = \frac{h}{e^2} \cdot \frac{1}{n}. \quad (3.47)$$

The same arguments hold equally for the IQHE and FQHE, in the former n is an integer, while n is a fraction for the latter. In FQHE, the interpretation is as follows: as we increase the flux from Φ to Φ_0 , a charge of magnitude e/m is transported from the inner circumference to the outer one when the flux is increased by Φ_0 units. The resultant Hall conductivity (or equivalently the resistivity) becomes,

$$\sigma_{xy} = \frac{e^2}{h} \cdot \frac{1}{m}. \quad (3.48)$$

Thus a whole electron is transferred only when the flux is increased by $m\Phi_0$ units.

3.2.8 Edge modes and conductivity of the single Landau level

When a particle is restricted to move only in one direction, the motion is said to be chiral where backscattering is prohibited. Thus the particles propagate in one direction at one edge of the sample and move in the other direction at the other end of the sample. Let us understand how the edge modes appear.

An edge can be modelled by a potential $V(x)$ in the y -direction which rises steeply, as shown in figure 3.10. Let us continue working in the Landau gauge, such that the Hamiltonian is given by,

$$\mathcal{H} = \frac{1}{2m} [p_x^2 + (p_y + eBx)^2] + V(x). \quad (3.49)$$

In the absence of the potential $V(x)$, the (lowest) wavefunction is a Gaussian of width $l_B (= \sqrt{\frac{\hbar}{eB}})$. If we assume that $V(x)$ is smooth over a distance l_B , and hence assume the center of each of Gaussian to be localized at $x = x_0$, we can Taylor expand the potential $V(x)$ about x_0 in the following fashion,

$$V(x) = V(x_0) + \frac{\partial V}{\partial x}(x - x_0) + \dots \quad (3.50)$$

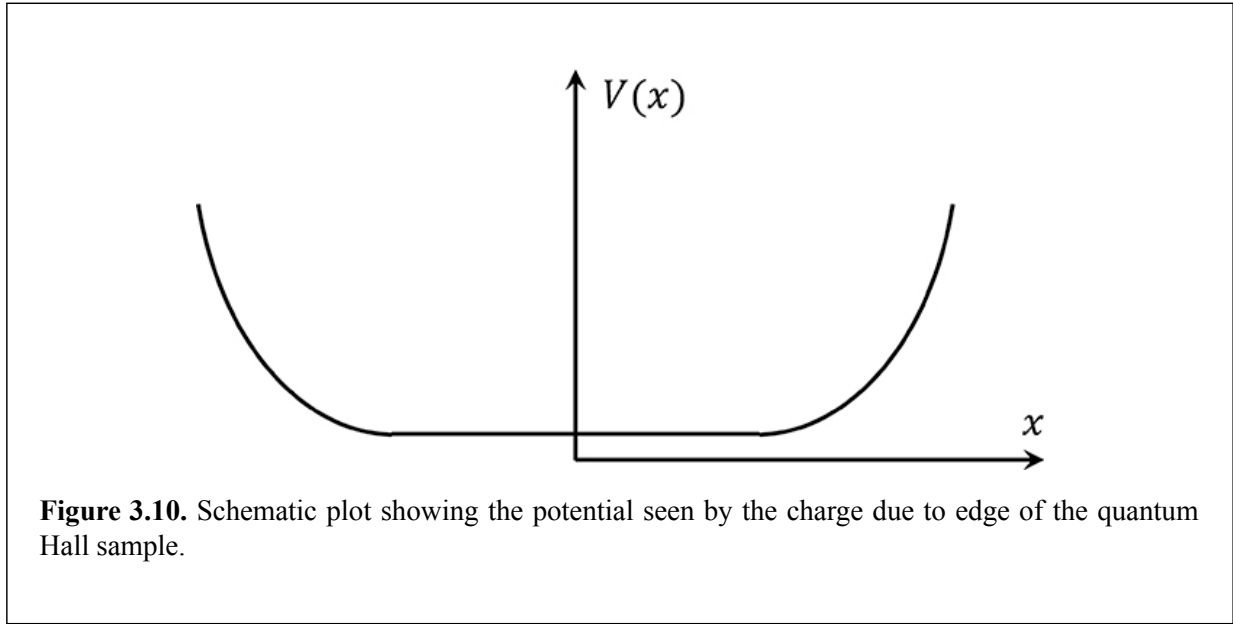
Dropping terms after the second one and assuming the constant term (first term) to be zero, the second term looks like the potential due to the electric field. So the particle acquires a drift velocity in the y -direction,

$$v_y = -\frac{1}{eB} \frac{\partial V}{\partial x}. \quad (3.51)$$

Each wavefunction labelled by a momentum, k is located at different x -positions, namely $x = -kl_B^2$, and thus has a different drift velocity. Additionally, the slopes of the two edges drawn above has different slopes, so $\frac{\partial V}{\partial x}$ has different signs at the edges. Thus v_y , at the left edge has a different sign with respect to the right edge. Further, because of the drift in y -direction, there will be a current I_y which is known as the Hall current and is calculated as,

$$\begin{aligned} I_y &= -e \int \frac{dk}{2\pi} v_y(k) \\ &= \frac{e}{2\pi l_B^2} \int dx \frac{1}{eB} \frac{dV}{dx}; \quad \text{using } l_B^2 = \frac{\hbar}{eB} \\ &= \frac{e^2}{2\pi\hbar} V_H \end{aligned} \tag{3.52}$$

V_H is the Hall voltage. Now, $\sigma_{xy} = \frac{I_y}{V_H} = \frac{e^2}{2\pi\hbar} = \frac{e^2}{h}$ which is indeed the expected conductivity for a single Landau level.



The above schematic diagram (figure 3.11) shows that the current is entirely carried by the edge states, since the bulk Landau level is absolutely flat (having no k dependence) and hence does not carry any current. The argument is also elegant as it does not depend upon the form of the potential $V(x)$.

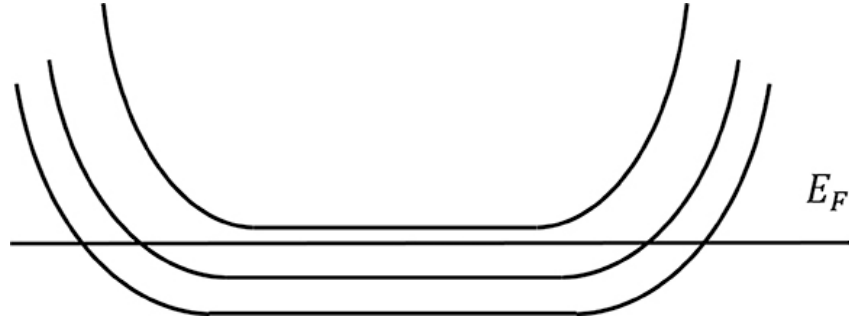


Figure 3.11. Schematic plot showing the appearance of edge modes.

Everything that we have discussed so far holds for a single Landau level, however the argument is equally valid for a large number of Landau levels, as long as the Fermi energy lies in between the filled and the unfilled Landau levels. Also, the chiral edge modes are robust to any impurity or disorder as there is no phase space available for scattering. If a left moving electron is to scatter over to a right moving electron, it has to cross the entire sample edge which is not allowed as the probability of scattering would be infinitesimally small owing to the macroscopic physical dimension of the sample. Thus the Hall plateaus are robust to disorder, defects and impurities.

A supremely important issue stands out: how do the plateaus exist in the first place? To see this, let us fix the electron density, n . Then we shall only have filled Landau levels when the magnetic field is exactly $B = \frac{n}{\nu} \Phi_0$ (with $\Phi_0 = \frac{h}{e}$) for some integer ν . But what happens when $B \neq \frac{n}{\nu} \Phi_0$, that is, when the Landau levels are partially filled? Also on top of that there is also a (small) electric field. In the partially filled last Landau level, the longitudinal conductivity will be non-zero, while the Hall conductivity will not be quantized. So how do the plateaus appear and the longitudinal conductivities vanish? The disorder comes to the rescue. It gives a finite width to the Landau levels. So even if the $B \neq \frac{n}{\nu} \Phi_0$, there is a finite regime over which the Hall conductivity, σ_{xy} remains constant and hence the plateaus form so that the Hall resistivity, ρ_{xy} freezes at that value, till the magnetic field is increased further.

3.2.9 Incompressibility and the QH states

The key feature of the QH states is that the states are incompressible. The compressibility, κ is defined by,

$$\kappa = -\frac{1}{A} \frac{\partial A}{\partial P} \Big|_N \quad (3.53)$$

where A , P and N are the area, pressure and the number of particles, respectively. The system is said to be incompressible when $\kappa = 0$, that is, when the area is insensitive to the pressure applied. The pressure is defined as the change of energy as a result of change of area, that is,

$$P = -\frac{\partial E}{\partial A}.$$

Thus the inverse of the compressibility is defined as,

$$\kappa^{-1} = -A \frac{\partial P}{\partial A} \big|_N = A \frac{\partial^2 E}{\partial A^2} \big|_N.$$

Since the energy is an extensive quantity, that is, it depends on the number of particles and hence can be written as,

$$E = N\epsilon(n)$$

where ϵ is energy per particle, and n is particle density, such that the total number of particles is defined by, $N = An$. Hence,

$$\kappa^{-1} = \frac{1}{n} = n^2 \left(2 \frac{d\epsilon(n)}{dn} + n \frac{d^2\epsilon(n)}{dn^2} \right) = n^2 \frac{d^2(n\epsilon)}{dn^2}. \quad (3.54)$$

Also, the chemical potential is given by,

$$\mu = \frac{\partial E}{\partial N} \big|_V = \frac{d(n\epsilon)}{dn}. \quad (3.55)$$

Thus comparing equations (3.54) and (3.55),

$$\kappa^{-1} = n^2 \frac{d\mu}{dn}. \quad (3.56)$$

The system is incompressible ($\kappa = 0$) when the chemical potential μ increases discontinuously as a function of density, that is, $\frac{\partial n}{\partial \mu} = 0$.

3.2.10 Hall effect in the symmetric gauge

The vector potential $\mathbf{A}$ in the mixed gauge can be written as (again yields $\mathbf{B} = B\hat{\mathbf{z}}$),

$$\mathbf{A} = -\frac{1}{2}(\mathbf{r} \times \mathbf{B}) = \frac{1}{2}(-By\hat{\mathbf{x}} + Bx\hat{\mathbf{y}}) \quad (3.57)$$

The choice of the symmetric gauge breaks the translation symmetry in both x - and y -directions. However, it preserves rotational symmetry about the origin. This of course means that angular momentum is a good quantum number. This is the most convenient gauge to study FQHE. Further, we can write down the non-canonical momentum as,

$$\boldsymbol{\pi} = \mathbf{p} + e\mathbf{A} = m\dot{\mathbf{r}}. \quad (3.58)$$

This is gauge invariant but non-canonical. One can use this to form the raising and lowering operators,

$$a = \frac{1}{\sqrt{\frac{eB\hbar}{\pi}}}(\pi_x - i\pi_y) = \sqrt{\frac{\pi}{eB}}(\pi_x - i\pi_y)$$

$$a^\dagger = \sqrt{\frac{\pi}{eB}}(\pi_x + i\pi_y) \quad (3.59)$$

where $2e\hbar B = 2e \frac{\hbar}{2\pi} B = \frac{eB\hbar}{\pi}$. Now,

$$\begin{aligned}
[a, a^\dagger] &= aa^\dagger - a^\dagger a = \left(\frac{\pi}{eB}\right)(\pi_x - i\pi_y)(\pi_x + i\pi_y) - (\pi_x + i\pi_y)(\pi_x - i\pi_y) \\
&= \left(\frac{\pi}{eB\hbar}\right)[\pi_x^2 + i\pi_x\pi_y - i\pi_y\pi_x + \pi_y^2 - \pi_x^2 - i\pi_x\pi_y + i\pi_y\pi_x - \pi_y^2] \\
&= \left(\frac{2\pi i}{eB\hbar}\right)[\pi_x\pi_y - \pi_y\pi_x] = \left(\frac{i}{eB\hbar}\right)[\pi_x, \pi_y]
\end{aligned} \tag{3.60}$$

π is not the canonical momenta in the sense, $[x_i, \pi_j] \neq \delta_{ij}$ and $[\pi_i, \pi_j] = \delta_{ij}$. However, they are gauge invariant. The numerical value of π does not depend upon the choice of gauge. This can be proved from the commutation relations,

$$\left\{p_i + eA_i, p_j + eA_j\right\} = -e\left(\frac{\partial A_j}{\partial x^i} - \frac{\partial A_i}{\partial x^j}\right) = -e\varepsilon_{ijk}B_k. \tag{3.61}$$

Thus, $[a, a^\dagger] = \left(\frac{i}{eB\hbar}\right)(-ie\hbar B) = 1$

To explore whether the Landau levels yield the expected degeneracy, we can introduce another momentum variable, namely,

$$\tilde{\pi} = \pi - e\mathbf{A}$$

$\tilde{\pi}$ is not gauge invariant, and depends on the gauge potential chosen. The vector potential, $\mathbf{A}$ enjoys the gauge freedom, that is,

$$\mathbf{A}' = \mathbf{A} - \nabla\chi$$

where χ is an arbitrary scalar. This yields,

$$\tilde{\pi}' = \mathbf{p}' - e(\mathbf{A}' - \nabla\chi) = \mathbf{p}' - e\mathbf{A}' + \nabla\chi.$$

The commutation relations for $\tilde{\pi}'$ are,

$$[\tilde{\pi}'_x, \tilde{\pi}'_y] = ie\hbar B \tag{3.62}$$

which differ only by a sign with respect to the π momenta. This is also an advantage of this new momenta in that they obey,

$$[\tilde{\pi}_i, \tilde{\pi}_j] = 0.$$

Finally, the Hamiltonian is written in terms of the a and $a^\dagger$ operators as,

$$\mathcal{H} = \hbar\omega_B\left(a^\dagger a + \frac{1}{2}\right) = \hbar\omega_B\left(n + \frac{1}{2}\right) \tag{3.63}$$

where $n = 0, 1, \dots$ and denote the indices of the Landau levels.

Now as a matter of exercise, in order to explore the degeneracy of the Landau levels, we introduce a second pair of raising and lowering operators, namely,

$$\begin{aligned}
b &= \frac{1}{\sqrt{2e\hbar B}}(\tilde{\pi}_x + i\tilde{\pi}_y) \\
b^\dagger &= \frac{1}{\sqrt{2e\hbar B}}(\tilde{\pi}_x - i\tilde{\pi}_y).
\end{aligned} \tag{3.64}$$

They too obey $[b, b^\dagger] = 1$. These $b, b^\dagger$ will yield the degeneracy of the Landau levels as shown in the following. Thus a general state in the Hilbert space $|n, m\rangle$ is defined by,

$$|n, m\rangle = \frac{(a^\dagger)^n (b^\dagger)^m}{\sqrt{n!m!}} |0, 0\rangle \quad (3.66)$$

where $a|0, 0\rangle = b|0, 0\rangle = 0$ and $\mathcal{H} = \frac{1}{2m} \boldsymbol{\pi} \cdot \boldsymbol{\pi} = \frac{1}{2m} (\mathbf{p} + e\mathbf{A})^2$.

Let us now construct the wavefunction in the symmetric gauge. we are going to focus on the lowest Landau level, $n = 0$ as it is of primary interest for discussing FQHE. The trick is to convert the definition of a into a differential equation,

$$\begin{aligned} a &= \frac{1}{\sqrt{2e\hbar B}} (\pi_x - i\pi_y) = \frac{1}{\sqrt{2e\hbar B}} [p_x - ip_y + e(A_x - iA_y)] \\ &= \frac{1}{\sqrt{2e\hbar B}} \left[-i\hbar \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) + \frac{eB}{2} (-y - ix) \right] \end{aligned} \quad (3.67)$$

using $z = x - iy$ and $\tilde{z} = x + iy$. Remember this is not usually how we define z and z^* (or $\tilde{z}$), however we shall stick to this definition. Also define,

$$\partial = \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right) \quad (3.68)$$

and,

$$\tilde{\partial} = \frac{1}{2} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) \quad (3.69)$$

which obey $\partial z = \tilde{\partial} \tilde{z} = 1$ and $\partial \tilde{z} = \tilde{\partial} z = 0$.

$$\partial z = \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right) (x - iy) = \frac{1}{2} (1 + 1) = 1.$$

So a and $a^\dagger$ in terms of the coordinates z can be written as,

$$a = -i\sqrt{2} \left(l_B \tilde{\partial} + \frac{z}{4l_B} \right); \quad a^\dagger = -i\sqrt{2} \left(l_B \tilde{\partial} - \frac{\tilde{z}}{4l_B} \right). \quad (3.70)$$

Now, the lowest Landau level is found by the one which is annihilated by this operator a .

$$a |0, m\rangle = 0$$

$$-i\sqrt{2} \left(l_B \tilde{\partial} + \frac{z}{4l_B} \right) |0, m\rangle = 0$$

$|0, m\rangle$ is called $\psi_{LLL}(z, \tilde{z})$, where LLL stands for lowest Landau level.

$$\psi_{LLL, m=0} \sim e^{-|z|^2/4l_B^2}$$

The ground state is known to be a Gaussian for a linear Harmonic oscillator⁶.

One can construct the higher Landau level wavefunctions by employing $b^\dagger$ successively to the $m = 0$ state. This yields,

$$\psi_{LLL,m} \sim \left(\frac{z}{l_B} \right)^m e^{-|z|^2/4l_B^2}. \quad (3.71)$$

It is straightforward to ascertain that $\psi_{LLL,m}$ are eigenfunctions of J_z , defined by,

$$J_z = \hbar(z\partial - \bar{z}\bar{\partial}) \quad (3.72)$$

and obey, $J_z\psi_{LLL,m} = m\hbar\psi_{LLL,m}$.

Let us explore the degeneracy associated with the Landau levels $\psi_{LLL,m}$, which is obtained by noting that the wavefunction with angular momentum, m is peaked on a circular ring of radius, $r = l_B\sqrt{2m}$. The number of states in an area, $A = \pi R^2$ is $\mathcal{N} = \pi R^2/\pi r^2 \simeq eBA/2\pi\hbar$ which is a result that we have seen earlier.

3.3 Kubo formula and the Hall conductivity

The important question at this stage is what protects the plateaus in the Hall conductivity (or the resistivity)? Why are they so flat and robust at the integer values (or at certain rational fractions)? Remember the system does not have either translational invariance (because of the presence of disorder) or time reversal invariance (because of the presence of a magnetic field). Thus two known symmetries are lost and still the plateaus persist⁷. We shall now show that the Hall conductivity assumes $\sigma_{xy} = \frac{\nu e^2}{h}$ (or equivalently $\rho_{xy} = \frac{h}{\nu e^2}$) under these conditions.

To compute the conductivity, we resort to the Kubo formula which arises from a more generalized concept, known as the ‘linear response theory’. We shall derive the Kubo formula under a few conditions for the sake of simplicity. They are,

- (i) Before the fields are applied, at $t = -\infty$, the system is in a non-interacting many-particle state which obeys $\mathcal{H}_0 |\psi_m\rangle = E_m |\psi_m\rangle$, where $\{E_m, \psi_m\}$ are the eigensolutions of $\mathcal{H}_0$.
- (ii) Even if we actually apply a constant electric field, $\mathbf{E}$, it is helpful to consider an alternating field with frequency, ω of the form, $\mathbf{E}(t) = \mathbf{E}e^{-i\omega t}$ and at the end of the calculation take the zero frequency limit, that is, $\omega \rightarrow 0$.
- (iii) Consider a gauge in which the transverse components of the vector potential are zero, that is, $A_t = 0$. In other words, $\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t}$ with no $\nabla\phi$ term, ϕ being the scalar potential. Equivalently one can assume that $\phi = \text{constant}$.

Now let us write the full Hamiltonian as,

$$\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_I \quad (3.73)$$

where $\mathcal{H}_0$ is the non-interacting Hamiltonian whose exact solutions are known (as described earlier) and $\mathcal{H}_I$ is the interaction term due to the coupling of the electrons to the external field. Thus $\mathcal{H}_I$ involves the current due to the motion of the electrons coupling with the vector potential arising due to the presence of the magnetic field. Thus,

$$\mathcal{H}_I = -\mathbf{J} \cdot \mathbf{A} \quad (3.74)$$

where $\mathbf{J}$ and $\mathbf{A}$ are the electric current density and the vector potential, respectively. $\mathbf{J}$ is related to the mechanical momentum $\mathbf{p} + e\mathbf{A}$. Using $\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t}$, one can write,

$$\mathbf{A} = \frac{\mathbf{E}}{i\omega} e^{-i\omega t}. \quad (3.75)$$

The aim is to compute the expectation value of the current density and find out how it depends on the applied electric field, such that the proportionality constant yields the conductivity. In particular we are interested in computing the Hall conductivity, σ_{xy} .

Here we shall consider the interaction picture where the time evolution of an arbitrary operator $\hat{O}$ is written as,

$$\hat{O}(t) = e^{i\mathcal{H}_0 t/\hbar} \hat{O}(0) e^{-i\mathcal{H}_0 t/\hbar}. \quad (3.76)$$

Here $\hat{O}$ can be any operator, such as $\mathbf{J}$ or $\mathcal{H}_I$. Further, the eigenstates in the interaction picture evolve with time according to,

$$|\psi(t)\rangle = U(t, t_0) |\psi(t_0)\rangle \quad (3.77)$$

where t_0 refers to an earlier time when the interaction is switched on and t denotes a later time. The time evolution operator, $U(t, t_0)$ is a unitary operator having a form,

$$U(t, t_0) = T \exp\left(-\frac{i}{\hbar} \int_{t_0}^t \mathcal{H}_I(t') dt'\right) \quad (3.78)$$

T denotes time ordering in the above equation [20]. If the interval $[t : t_0]$ is split into several time steps, T keeps the earliest time to the right. Now let us consider that as $t_0 \rightarrow -\infty$, that is, before the perturbation is switched on, the system was in the ground state $|\psi_0(t)\rangle$. Hence the time evolution operator can be written as,

$$U(t, t_0 = -\infty) = T \exp\left(-\frac{i}{\hbar} \int_{-\infty}^t \mathcal{H}_I(t') dt'\right) = U(t) \quad \text{say.} \quad (3.79)$$

The ground state expectation value of the current operator is given by,

$$\begin{aligned} \langle \mathbf{J}(t) \rangle &= \langle \psi_0(t) | \mathbf{J}(t) | \psi_0(t) \rangle = \langle \psi_0 | U^{-1}(t) \mathbf{J}(t) U(t) | \psi_0 \rangle \\ &= \langle \psi_0 | \left[T e^{\frac{i}{\hbar} \int_{-\infty}^t \mathcal{H}_I(t') dt'} \mathbf{J}(t) T e^{-\frac{i}{\hbar} \int_{-\infty}^t \mathcal{H}_I(t') dt'} \right] | \psi_0 \rangle. \end{aligned} \quad (3.80)$$

An expansion of the exponentials (assuming the interaction term to be weak) and retaining terms up to first order in $\mathcal{H}_I$ yields,

$$\langle \mathbf{J}(t) \rangle \approx \langle \psi_0 | \left[\mathbf{J}(t) + \frac{i}{\hbar} \int_{-\infty}^t dt' [\mathcal{H}_I(t'), \mathbf{J}(t)] \right] | \psi_0 \rangle. \quad (3.81)$$

The second term inside the bracket of the RHS involves a commutator of $\mathcal{H}_I$ and $\mathbf{J}$. It is to be kept in mind that the commutator does not vanish at two arbitrary times t and t' . Using $\mathbf{A}(t) = \frac{\mathbf{E}}{i\omega} e^{-i\omega t}$ and $\mathcal{H}_I(t) = -\mathbf{J} \cdot \mathbf{A}$, the $\langle \mathbf{J}(t) \rangle$ takes the form,

$$\langle \mathbf{J}(t) \rangle \approx \langle \psi_0 | \left[\mathbf{J}(t) + \frac{i}{\hbar} \int_{-\infty}^t dt' [-\mathbf{J}(t') \cdot \frac{\mathbf{E}}{i\omega} e^{-i\omega t'}, \mathbf{J}(t)] \right] | \psi_0 \rangle. \quad (3.82)$$

The first term inside the bracket in the RHS is the current due to the absence of an external electric field which can safely be ignored in our case. So only the second term survives. Thus, writing for components α, β ($\alpha, \beta \in x, y, z$)

$$\langle J_\alpha(t) \rangle = \frac{1}{\hbar\omega} \int_{-\infty}^t dt' \langle \psi_0 | [J_\beta(t'), J_\alpha(t)] | \psi_0 \rangle E_\beta e^{-i\omega t'} \quad (3.83)$$

where $\mathbf{J} \cdot \mathbf{E}$ is written as $J_\beta E_\beta$.

Since the system is invariant under time translation, the above correlation depends on $t - t'$ and not on t and t' individually. Introducing a new variable $t - t' = \tilde{t}$,

$$\langle J_\alpha(t) \rangle = \frac{1}{\hbar\omega} \left(\int_0^\infty d\tilde{t} e^{i\omega\tilde{t}} \langle \psi_0 | [J_\beta(0), J_\alpha(\tilde{t})] | \psi_0 \rangle \right) E_\beta e^{-i\omega t} \quad (3.84)$$

where the term inside the bracket in the RHS can be written as, $\sigma_{\alpha\beta} E_\beta e^{-i\omega t}$, $\sigma_{\alpha\beta}$ being the components of the conductivity tensor. Note that the time t dependence is outside the integral and appears as $e^{-i\omega t}$. Thus in the linear response regime, if an electric field of frequency, ω is applied, the current responds to the external field by oscillating with the same frequency as the external field.

The Hall conductivity is the off-diagonal component, and can be computed using,

$$\sigma_{xy}(\omega) = \frac{1}{\hbar\omega} \int_0^\infty dt e^{i\omega t} \langle \psi_0 | [J_y(0), J_x(t)] | \psi_0 \rangle \quad (3.85)$$

$J_x(t)$ can be written as,

$$J_x(t) = e^{i\mathcal{H}_0 t/\hbar} J_x(0) e^{-i\mathcal{H}_0 t/\hbar} \quad (3.86)$$

in the above expression, and using completeness relation of the states, namely, $|\psi_n\rangle\langle\psi_n| = \mathbf{1}$, one obtains,

$$\begin{aligned} \sigma_{xy}(\omega) = \frac{1}{\hbar\omega} \int_0^\infty dt e^{i\omega t} \sum_n \langle \psi_0 | J_y | \psi_n \rangle \langle \psi_n | J_x | \psi_0 \rangle e^{i(E_n - E_0)t/\hbar} \\ - \langle \psi_0 | J_x | \psi_n \rangle \langle \psi_n | J_y | \psi_0 \rangle e^{i(E_0 - E_n)t/\hbar}. \end{aligned} \quad (3.87)$$

Now we shall perform the integral over time, t . Let us write $\frac{E_n}{\hbar} = \omega_n$ and introduce $\omega \rightarrow \omega + i\epsilon$ where ϵ is infinitesimal quantity considered for the convergence of the integral. Then the Hall conductivity is obtained as,

$$\sigma_{xy}(\omega) = -\frac{i}{\omega} \sum_{E_n \neq E_0} \left[\frac{\langle \psi_0 | J_y | \psi_n \rangle \langle \psi_n | J_x | \psi_0 \rangle}{\hbar\omega + E_n - E_0} - \frac{\langle \psi_0 | J_x | \psi_n \rangle \langle \psi_n | J_y | \psi_0 \rangle}{\hbar\omega + E_0 - E_n} \right]. \quad (3.88)$$

Finally, we shall take the limit $\omega \rightarrow 0$ to account for the constant field. For that let us expand the denominator as follows,

$$\begin{aligned}\frac{1}{\hbar\omega + E_n - E_0} &\approx \frac{1}{E_n - E_0} - \frac{\hbar\omega}{(E_n - E_0)^2} + O(\omega^2) \\ \frac{1}{\hbar\omega + E_0 - E_n} &\approx \frac{1}{E_0 - E_n} - \frac{\hbar\omega}{(E_0 - E_n)^2}.\end{aligned}\quad (3.89)$$

The first term looks divergent and such divergence is responsible for the peak in the longitudinal conductivity which, by now we are familiar with. Moreover, $\sigma_{xy}(\omega)$ should not contain a term which is independent of ω . This is the DC conductivity ($\omega = 0$) which is absent in a translationally invariant system. Thus finally one arrives at,

$$\sigma_{xy}(\omega) = i\hbar \sum_{n \neq 0} \frac{\langle \psi_0 | J_y | \psi_n \rangle \langle \psi_n | J_x | \psi_0 \rangle - \langle \psi_0 | J_x | \psi_n \rangle \langle \psi_n | J_y | \psi_0 \rangle}{(E_n - E_0)^2}.$$
(3.90)

This is the Kubo formula for the Hall conductivity.

To proceed further, let us assume a specific case of perturbing a system below [21]. Consider a quantum Hall sample in the form of a torus (or a donut). Let us thread two fluxes Φ_x and Φ_y (instead of one) as shown in figure 3.12. Owing to this, the gauge potentials can be written as,

$$A_x = \frac{\Phi_x}{L_x}, A_y = \frac{\Phi_y}{L_y} + Bx.$$
(3.91)

It is clear that the states of the quantum system are sensitive to non-integer values of Φ_i/Φ_0 ($i = x, y$) where, $\Phi_0 = \frac{h}{e}$. Specifically, if we increase either Φ_x or Φ_y from 0 to Φ_0 then the spectrum of the quantum system must remain invariant. Hence writing the perturbation Hamiltonian in terms of the fluxes,

$$\mathcal{H}' = - \sum_{i \in 1,2} \frac{J_i \Phi_i}{L_i}.$$
(3.92)

To the first order corresponding to this perturbation term, the modified ground state becomes,

$$| \psi'_0 \rangle = | \psi_0 \rangle + \sum_{\substack{\psi_n \neq \psi_0 \\ E_n \neq E_0}} \frac{\langle \psi_n | \mathcal{H}' | \psi_0 \rangle}{E_n - E_0} | \psi_n \rangle.$$
(3.93)

Thus writing,

$$\Delta | \psi \rangle = | \psi'_0 \rangle - | \psi_0 \rangle = \sum_{\substack{\psi_n \neq \psi_0 \\ E_n \neq E_0}} \frac{\langle \psi_n | \mathcal{H}' | \psi_0 \rangle}{E_n - E_0} | \psi_n \rangle.$$
(3.94)

Considering infinitesimal changes in Φ_i we can write,

$$\left| \frac{\partial | \Delta \psi \rangle}{\partial \Phi_i} \right\rangle = -\frac{1}{L_i} \sum_n \frac{\langle \psi_n | J_i | \psi_0 \rangle}{E_n - E_0} | \psi_n \rangle.$$
(3.95)

Terms like those in the RHS of the above equation appeared in the Hall conductivity. Let us write the total Hall conductivity including the area factor $L_x L_y$ of the sample, which can be written as,

$$\begin{aligned}
\sigma_{xy} &= i\hbar L_x L_y \sum_{\psi_n \neq \psi_0, E_n \neq E_0} \frac{\langle \psi_0 | J_y | \psi_n \rangle \langle \psi_n | J_x | \psi_0 \rangle - \langle \psi_0 | J_x | \psi_n \rangle \langle \psi_n | J_y | \psi_0 \rangle}{(E_n - E_0)^2} \\
&= i\hbar \left[\left\langle \frac{\partial \psi_0}{\partial \Phi_y} \middle| \frac{\partial \psi_0}{\partial \Phi_x} \right\rangle - \left\langle \frac{\partial \psi_0}{\partial \Phi_x} \middle| \frac{\partial \psi_0}{\partial \Phi_y} \right\rangle \right] = i\hbar \left[\frac{\partial}{\partial \Phi_y} \left\langle \psi_0 \middle| \frac{\partial \psi_0}{\partial \Phi_x} \right\rangle - \frac{\partial}{\partial \Phi_x} \left\langle \psi_0 \middle| \frac{\partial \psi_0}{\partial \Phi_y} \right\rangle \right] \quad (3.96)
\end{aligned}$$

$\left\langle \psi_0 \middle| \frac{\partial \psi_0}{\partial \Phi_x \partial \Phi_y} \right\rangle$ will cancel from both the terms This is the expression for Hall conductivity.

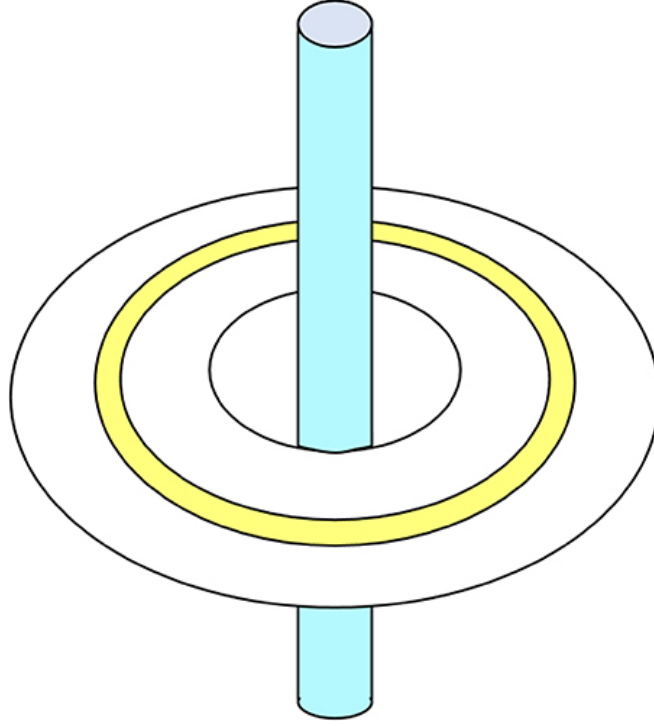


Figure 3.12. The schematic plot shows a disc threaded by a flux perpendicular to the plane and another one in the plane of the disc.

3.3.1 Hall conductivity and the Chern number

Remember the spectrum of the Hamiltonian depends upon $\Phi_i \bmod \Phi_0$ ⁸. If there is a remainder, then the division does not yield an integer, and if there is none, then $\Phi_i/\Phi_0 = \text{integer}$. Φ_i being parameters of $\mathcal{H}$, Φ_i 's are periodic functions. To emphasize the periodicity, we shall introduce angular variables, θ_i such that,

$$\theta_i = \frac{2\pi\Phi_i}{\Phi_0} \quad \text{where } \theta_i \in [0, 2\pi]. \quad (3.97)$$

As θ_i increases from 0 to 2π , Φ_i increases from $0 \rightarrow \Phi_0$. Now rewrite $\frac{\partial}{\partial \Phi_i}$ as $\frac{\partial}{\partial \theta_i}$ and introduce a quantity called Berry connection which is defined on the surface of the torus as,

$$\mathcal{A}_i(\Phi) = -i \langle \psi_0 | \frac{\partial}{\partial \theta_i} | \psi_0 \rangle. \quad (3.98)$$

Further, we define a quantity called the Berry curvature (analogous to the magnetic field),

$$\mathcal{F}_{xy} = \frac{\partial \mathcal{A}_x}{\partial \theta_y} - \frac{\partial \mathcal{A}_y}{\partial \theta_x} = (\nabla_\theta \times \mathcal{A}) = -i \left[\frac{\partial}{\partial \theta_y} \left\langle \psi_0 \left| \frac{\partial \psi_0}{\partial \theta_x} \right\rangle - \frac{\partial}{\partial \theta_x} \left\langle \psi_0 \left| \frac{\partial \psi_0}{\partial \theta_y} \right\rangle \right]. \quad (3.99)$$

Note that the last term above in equation (3.99) is the Hall conductivity which is formally written in terms of the Berry curvature as,

$$\sigma_{xy} = -\frac{e^2}{h} \mathcal{F}_{xy}. \quad (3.10)$$

We are still left with the task of understanding the quantization of σ_{xy} which is central to the discussion on QHE. We can now integrate over the surface of the torus to get the total conductivity,

$$\sigma_{xy} = -\frac{e^2}{h} \int_{\text{torus}} \frac{d^2 \theta}{(2\pi)^2} \mathcal{F}_{xy}. \quad (3.10)$$

The quantity $C = \frac{1}{2\pi} \int d^2 \theta \mathcal{F}_{xy}$ is called the first Chern number. Thus if we average over the fluxes the conductivity assumes a form,

$$\sigma_{xy} = -\frac{e^2}{h} C \quad (3.10)$$

C is necessarily an integer. It is also referred to as the TKNN invariant after Thouless, Kohmoto, Nightingale and den Nijs [22].

Here we provide an argument that the Chern number which is the integral over the Berry curvature is indeed an integer. For simplicity, let us assume a translationally invariant system in which the eigenstates can be represented by the Bloch functions. That is, $|\psi\rangle_0$ appearing above can be written as $u_k e^{i\phi_k}$ where u_k captures the periodicity of the lattice. Since the Berry connection requires a derivative to be taken, namely, $\langle \psi_0 | \frac{\partial}{\partial \theta} | \psi_0 \rangle$, which would be equivalent to,

$$\frac{\partial}{\partial \phi_k} (u_k e^{i\phi_k}) \approx \nabla_k \phi_k. \quad (3.10)$$

Now when one takes an integral over the Brillouin zone (which is equivalent to the surface of the torus in real space), then $(\nabla_k \phi_k) \cdot dk$ over a closed surface is zero.

$$\oint \nabla_k \cdot \phi_k dk = 0. \quad (3.10)$$

This says that the measurable quantity, $e^{i\phi}$ obeys⁹,

$$e^{i\phi(0)} = e^{i\phi(2\pi)}. \quad (3.10)$$

Thus,

$$\begin{aligned} |\phi(0) - \phi(2\pi)| &= 2\pi \times (\text{some integer}) \\ &= 2\pi C. \end{aligned}$$

Thus the integral over the curvature, (3.10
6)

$$\oint \mathcal{F} \frac{d^2 k}{2\pi} = \oint \nabla_k \phi_k dk = \frac{2\pi C}{2\pi} = C. \quad (3.10)$$

Thus Chern number is an integer and hence we get the Hall conductivity to be quantized in unit of e^2/h . The above calculations are of course applicable to IQHE. 7)

3.4 Quantum Hall effect in graphene

Having studied the quantized Hall effect in a 2DEG in detail, we focus on another system which is of topical interest, namely, graphene. Apart from reviewing the basic electronic properties of graphene, we discuss the properties of the Landau levels. The unequal spacing between the successive Landau levels is a feature that shows up in many experiments. We discuss some of them.

Graphene is made of a single layer of carbon (C) atoms arranged in a honeycomb lattice structure. Each C atom has six electrons with an electronic configuration $1s^2 2s^2 2p^2$. Four electrons in the $2s$ and the $2p$ orbitals create a hybridized sp^2 bonding orbital. The orbital connects to the three nearest neighbours in the plane of the crystal lattice, while the fourth electron occupies a π orbital that projects out above and below the plane. This π orbital has a significant overlap with those from the neighbouring C atoms. This is responsible for rendering excellent mobility of graphene, while the σ bonds that connect a C atom to its three neighbours yield the stability of the crystal structure. The discovery of such one-atom thick planar material (graphene is a perfect example of a 2D material realized so far) earned a Nobel prize for A Geim and K Novoselov, both from the University of Manchester in UK in 2010. Before we embark on the Hall effect in graphene, let us review the electronic properties.

3.4.1 Basic electronic properties of graphene

Owing to the large mobility of π electrons in graphene, a nearest neighbour tight binding Hamiltonian of the following form is most suitable, namely,

$$\mathcal{H} = -t \sum_{\langle ij \rangle, \sigma} (a_{i\sigma}^\dagger b_{j\sigma} + \text{h. c.}) \quad (3.10)$$

where $a_i^\dagger(b_i)$ denote creation (annihilation) operators for electrons at the A(B) sublattice sites. σ 8) denotes the spin of the electrons, however, that will be suppressed in the next step onwards owing to no active role played by the spin either in the band structure or in QHE. We shall make the spin degrees of freedom apparent only when it is needed. Here $t \simeq 2.7$ eV which is considerably large and allows us to ignore electron–electron interaction. The vectors connecting the nearest neighbours, δ_i , direct, $\mathbf{a}_i$ and the reciprocal $\mathbf{b}_i$ lattice vectors (see figure 3.13) are written as,

$$\begin{aligned} \delta_1 &= \frac{a}{2} (\sqrt{3}\hat{x} + \hat{y}); & \delta_2 &= \frac{a}{2} (-\sqrt{3}\hat{x} + \hat{y}); & \delta_3 &= -a\hat{x} \\ \mathbf{a}_1 &= \frac{a}{2} (\sqrt{3}\hat{x} + 3\hat{y}); & \mathbf{a}_2 &= \frac{a}{2} (-\sqrt{3}\hat{x} + 3\hat{y}) \\ \mathbf{b}_1 &= \frac{2\pi}{3a} (\sqrt{3}\hat{k}_x + \hat{k}_y); & \mathbf{b}_2 &= \frac{2\pi}{3a} (-\sqrt{3}\hat{k}_x + 3\hat{k}_y) \end{aligned} \quad (3.10 \quad 9)$$

with the lattice constant, $a = 1.42\text{\AA}$. Using the nearest neighbour vectors, δ_i , we explicitly write the tight binding Hamiltonian as,

$$\mathcal{H} = -t \sum_{\mathbf{R}, \delta} [b^\dagger(\mathbf{R} + \delta_i) a(\mathbf{R}) + a^\dagger(\mathbf{R}) b(\mathbf{R} + \delta_i)]. \quad (3.11 \quad 0)$$

The lattice vector $\mathbf{R}$ at an arbitrary site is given by,

$$\mathbf{R} = n\mathbf{a}_1 + m\mathbf{a}_2 \quad n, m \in N. \quad (3.11 \quad 1)$$

The Fourier transform for these operators is done using,

$$a_{\mathbf{k}} = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} e^{-i\mathbf{k} \cdot \mathbf{R}} a(\mathbf{R}). \quad (3.11 \quad 2)$$

This yields the Hamiltonian in the momentum space as follows,

$$\mathcal{H} = -\frac{t}{N} \sum_{\mathbf{k}, \mathbf{q}} \sum_{\mathbf{R}} \sum_{i=1}^3 e^{i(\mathbf{k}-\mathbf{q}) \cdot \mathbf{R}} \left[e^{i\mathbf{q} \cdot \delta_i} b_{\mathbf{q}}^\dagger a_{\mathbf{k}} + e^{i(\mathbf{k}-\mathbf{q}) \cdot \mathbf{R}} e^{-i\mathbf{q} \cdot \delta_i} a_{\mathbf{q}}^\dagger b_{\mathbf{k}} \right]. \quad (3.11 \quad 3)$$

Using the following definition of the Kronecker delta,

$$\delta_{\mathbf{k}, \mathbf{q}} = \frac{1}{N} \sum_{\mathbf{R}} e^{i(\mathbf{k}-\mathbf{q}) \cdot \mathbf{R}} \quad (3.11 \quad 4)$$

one gets,

$$\begin{aligned} \mathcal{H} &= -t \sum_{\mathbf{k}} \sum_{i=1}^3 \left[e^{-i\mathbf{k} \cdot \delta_i} b_{\mathbf{k}}^\dagger a_{\mathbf{k}} + e^{i\mathbf{k} \cdot \delta_i} a_{\mathbf{k}}^\dagger b_{\mathbf{k}} \right] \\ &= -t \sum_{\mathbf{k}} \sum_{i=1}^3 \begin{pmatrix} a_{\mathbf{k}}^\dagger & b_{\mathbf{k}}^\dagger \end{pmatrix} \begin{pmatrix} 0 & e^{-i\mathbf{k} \cdot \delta_i} \\ e^{i\mathbf{k} \cdot \delta_i} & 0 \end{pmatrix} \begin{pmatrix} a_{\mathbf{k}} \\ b_{\mathbf{k}} \end{pmatrix} \\ &= -t \sum_{\mathbf{k}} \sum_{i=1}^3 \begin{pmatrix} a_{\mathbf{k}}^\dagger & b_{\mathbf{k}}^\dagger \end{pmatrix} h(\mathbf{k}) \begin{pmatrix} a_{\mathbf{k}} \\ b_{\mathbf{k}} \end{pmatrix} \end{aligned} \quad (3.11 \quad 5)$$

where $h(\mathbf{k})$ is the Hamiltonian matrix defined by,

$$h(\mathbf{k}) = -t \begin{pmatrix} 0 & (e^{i\mathbf{k} \cdot \delta_1} + e^{i\mathbf{k} \cdot \delta_2} + e^{i\mathbf{k} \cdot \delta_3}) \\ (e^{-i\mathbf{k} \cdot \delta_1} + e^{-i\mathbf{k} \cdot \delta_2} + e^{-i\mathbf{k} \cdot \delta_3}) & 0 \end{pmatrix}. \quad (3.11 \quad 6)$$

Since the difference between two nearest neighbour lattice vectors, δ_i and δ_j must yield a lattice vector, $\mathbf{R}$, we can do a transformation,

$$a_{\mathbf{k}} \rightarrow e^{i\mathbf{k} \cdot \delta_3} a_{\mathbf{k}}, \quad \text{and} \quad a_{\mathbf{k}}^\dagger \rightarrow e^{-i\mathbf{k} \cdot \delta_3} a_{\mathbf{k}}^\dagger. \quad (3.11 \quad 7)$$

The above transformation yields a new Hamiltonian matrix,

$$\tilde{h}(\mathbf{k}) = -t \begin{pmatrix} 0 & (e^{i\mathbf{k} \cdot \delta_1} + e^{i\mathbf{k} \cdot \delta_2} + 1) \\ (e^{-i\mathbf{k} \cdot \delta_1} + e^{-i\mathbf{k} \cdot \delta_2} + 1) & 0 \end{pmatrix}. \quad (3.11)$$

Using the definitions of δ_i , one gets,

$$\tilde{h}(\mathbf{k}) = -t \begin{pmatrix} 0 & -(e^{i\mathbf{k} \cdot \mathbf{a}_1} + e^{i\mathbf{k} \cdot \mathbf{a}_2} + 1) \\ -(e^{-i\mathbf{k} \cdot \mathbf{a}_1} + e^{-i\mathbf{k} \cdot \mathbf{a}_2} + 1) & 0 \end{pmatrix}. \quad (3.11)$$

One can check that $\tilde{h}(\mathbf{k})$ obeys $\tilde{h}(\mathbf{k}) = \tilde{h}(\mathbf{k} + \mathbf{G})$, where $\mathbf{G}$ is the reciprocal lattice vector, defined as, $\mathbf{G} = p\mathbf{b}_1 + q\mathbf{b}_2$, with p and q being integers. Thus,

$$\tilde{h}(\mathbf{k}) = -t \begin{pmatrix} 0 & f(\mathbf{k}) \\ f^*(\mathbf{k}) & 0 \end{pmatrix} \quad (3.12)$$

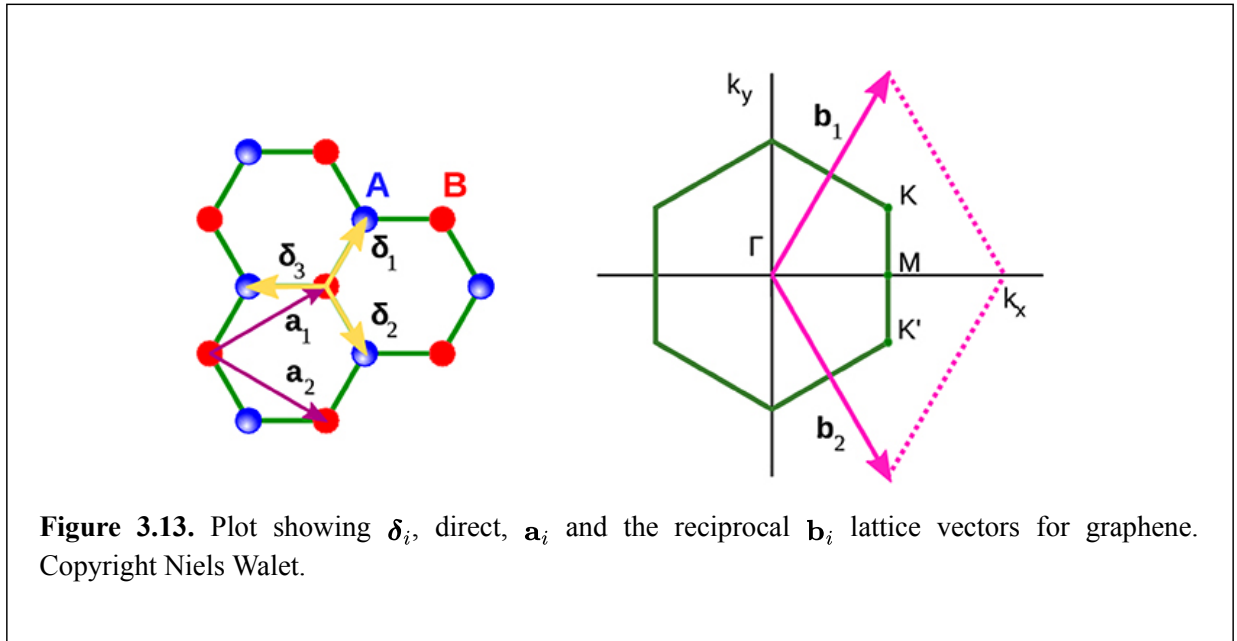
where

$$f(\mathbf{k}) = -t \left(e^{-ik_x a} + 2e^{ik_x a/2} \cos\left(\frac{k_y \sqrt{3}a}{2}\right) \right).$$

The tight binding energy is obtained by diagonalizing $\tilde{h}(\mathbf{k})$ which yields,

$$\epsilon_{\mathbf{k}} = \pm t \sqrt{3 + 2 \cos(\sqrt{3}ak_y) + 4 \cos(\sqrt{3}ak_y/2) \cos(3ak_x/2)}. \quad (3.12)$$

The two bands described by the '+' and the '-' signs in the above dispersion touch at six points in the Brillouin zone. Since graphene has one accessible electron per C atom, one can assume a half-filled system where the lower band is completely filled. Further, we wish to discuss the low lying excitations just above the ground state of the system.



This necessitates exploring the low energy theory of graphene. To achieve that we have to identify the band touching points which can be obtained from the condition, $f(\mathbf{k}) = 0$. Separately putting the real and the imaginary parts equal to zero yield,

$$\begin{aligned} \cos(k_x a) + 2 \cos(k_x a/2) \cos(\sqrt{3}k_y a/2) &= 0 \\ -\sin(k_x a) + 2 \sin(k_x a/2) \cos(\sqrt{3}k_y a/2) &= 0. \end{aligned} \quad (3.12 \quad 2)$$

equation (3.122) can be manipulated as follows,

$$\sin(k_x a/2) \left[-\cos(k_x a/2) + \cos(k_y a\sqrt{3}/2) \right] = 0. \quad (3.12 \quad 3)$$

Thus one is left with two options, namely,

$$\begin{aligned} \text{either } (i) \quad \sin(k_x a/2) &= 0; \quad \text{which means} \quad \cos(k_x a/2) = \pm 1; \\ \text{or, } (ii) \quad \cos(k_x a/2) &= \cos(\sqrt{3}k_y a/2). \end{aligned}$$

Option (i) gives us,

$$1 + 2 \cos(k_y \sqrt{3}a/2) = 0$$

which yields the points $\left(0, \pm \frac{4\pi}{3\sqrt{3}a}\right)$ (plus or minus the reciprocal lattice vector, $\mathbf{G}$), whereas, option (ii) can be written as,

$$\cos(k_y a\sqrt{3}) + 2 \cos^2(k_y a\sqrt{3}/2) = 0.$$

Thus we get four more points, which are, $\pm \frac{2\pi}{3a} \left(1, \frac{1}{\sqrt{3}}\right)$, and $\pm \frac{2\pi}{3a} \left(1, -\frac{1}{\sqrt{3}}\right)$ (again plus or minus the reciprocal lattice vector, $\mathbf{G}$).

A closer inspection yields all the six points are not independent. For example, the set of vectors, namely, $\left(0, \pm \frac{4\pi}{3\sqrt{3}a}\right)$, $\frac{2\pi}{3a} \left(1, -\frac{1}{\sqrt{3}}\right)$ and $\frac{2\pi}{3a} \left(-1, -\frac{1}{\sqrt{3}}\right)$ can be connected to each other via the combination of the reciprocal lattice vectors, $\mathbf{b}_1$ and $\mathbf{b}_2$. For example,

$$\begin{aligned} \left(0, \frac{4\pi}{3\sqrt{3}a}\right) + \mathbf{b}_2 &= \frac{2\pi}{3a} \left(1, -\frac{1}{\sqrt{3}}\right) \\ \left(0, \frac{4\pi}{3\sqrt{3}a}\right) - \mathbf{b}_1 &= \frac{2\pi}{3a} \left(-1, -\frac{1}{\sqrt{3}}\right). \end{aligned} \quad (3.12 \quad 4)$$

The same is true for the other vectors,

$$\left(0, \frac{4\pi}{3\sqrt{3}a}\right), \quad \frac{2\pi}{3a} \left(-1, \frac{1}{\sqrt{3}}\right), \quad \frac{2\pi}{3a} \left(1, \frac{1}{\sqrt{3}}\right).$$

Thus, only two of them are found to be independent. Traditionally, they are called $\mathbf{K}$ and $\mathbf{K}'$ and can be written as,

$$\mathbf{K} = \frac{2\pi}{3a} \left(1, \frac{1}{\sqrt{3}}\right), \quad \text{and} \quad \mathbf{K}' = \frac{2\pi}{3a} \left(1, -\frac{1}{\sqrt{3}}\right).$$

Any other independent pair is also a valid choice for $\mathbf{K}$ and $\mathbf{K}'$.

It may be kept in mind that the two bands touch at these points and the gap between the conduction and the valence band closes. Thus there are two branches of low energy excitations, namely one of them with momentum close to $\mathbf{K}$ and the other close to $\mathbf{K}'$. Since $f(\mathbf{k})$ becomes zero at $\mathbf{k} = \mathbf{K}$. Defining $\mathbf{q} = \mathbf{k} - \mathbf{K}$, one can expand $f(\mathbf{k})$ near $\mathbf{K}$ in Taylor series about $\mathbf{q} = 0$,

$$\begin{aligned} f'(\mathbf{q}) &= \frac{\partial f(\mathbf{k})}{\partial k_x} \big|_{(k_x=K_x)} (k_x - K_x) + \frac{\partial f(\mathbf{k})}{\partial k_y} \big|_{(k_y=K_y)} (k_y - K_y) \\ &= \frac{3at}{2} (q_x + iq_y). \end{aligned}$$

Thus the energy spectrum assumes the form,

$$\epsilon_{\mathbf{K}}(\mathbf{q}) = \hbar v_F (q_x + iq_y) \quad (3.12)$$

where v_F is the Fermi velocity defined by, $v_F = \frac{3at}{2\hbar} \simeq 10^6 \text{ ms}^{-1}$. Similarly, if we expand around $\mathbf{K}'$, one gets,

$$\epsilon_{\mathbf{K}'}(\mathbf{q}) = \hbar v_F (q_x - iq_y). \quad (3.12)$$

Thus in a general notation, we can write,

$$\epsilon_{\mathbf{K},\mathbf{K}'} = \hbar v_F \mathbf{q} \cdot \boldsymbol{\sigma} \quad (3.12)$$

where $\mathbf{q}$ is a planar vector (q_x, q_y) and $\boldsymbol{\sigma}$ is the Pauli matrix vector (σ_x, σ_y) . The electrons close to the $\mathbf{K}$ and $\mathbf{K}'$ points are called massless Dirac fermions, as they obey the Dirac equation without the ‘mass’ term¹⁰. It may be noted that,

$$\epsilon_{\mathbf{K}'}(\mathbf{q}) = \epsilon_{\mathbf{K}}^*(\mathbf{q}). \quad (3.12)$$

This implies that (as will be seen later) the ‘helicity’ of the electrons is opposite at $\mathbf{K}'$ with respect to $\mathbf{K}$.

To sum up our preliminary discussion on graphene, we note that the low energy properties are governed by the dispersion,

$$\epsilon(q) \simeq \pm v_F |q| \quad (3.12)$$

which implies that the eigenvalues are only functions of the magnitude of the wavevector $\mathbf{q}$, and does not depend upon its direction in the 2D plane. The corresponding linear dispersion is shown in figure 3.14. Also the Hamiltonian on a formal note denotes that of a massless $s = 1/2$ particle, such as a neutrino, however the velocity of the particles is reduced by a factor of 300 compared to the speed of light. Further, the handedness (or the helicity) feature of neutrinos is in-built, where the electrons behave similar to a ‘left handed’ neutrino at the Dirac point $\mathbf{K}$ and as a ‘right handed’ neutrino at $\mathbf{K}'$ or vice versa.

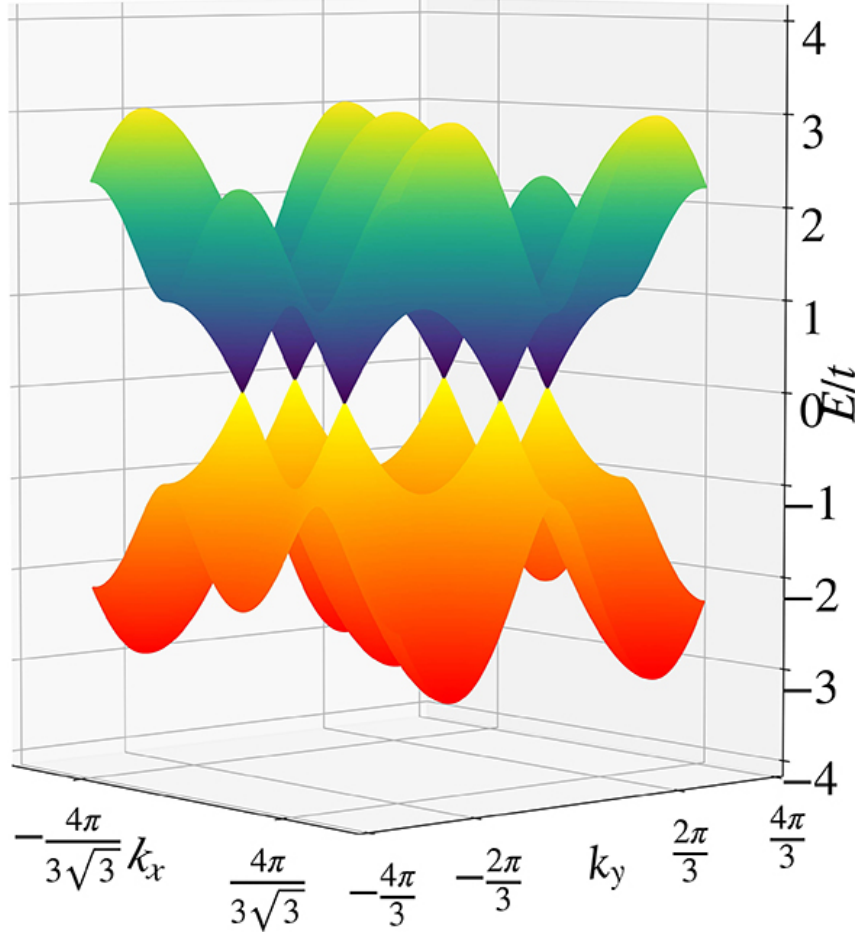


Figure 3.14. Plot showing the two tight binding bands of graphene in the first BZ. The two bands touch at six points. In the vicinity of those points, the bands are linearly dispersing bearing signatures of (pseudo-)relativistic physics.

3.4.2 Experimental confirmation of the Dirac spectrum

When a beam of monochromatic photons with an energy larger than the work function of a particular material interacts with the constituent charges (electrons) by indenting on the surface of the sample, the electrons absorb the photons and thus possess sufficient energy to escape from the sample. By measuring the energy and the momentum of the photoelectrons and using energy–momentum conservations laws, one can derive the properties of the electrons prior to them being incident on the surface and relate them with those getting scattered. Angle resolved photoemission spectroscopy (ARPES) can be a direct probe to resolve the momentum dependent band structure and the topology of the Fermi surface. In ARPES, a photon is employed to eject an electron from the surface of the graphene layer. The intensity of the ARPES is proportional to the transition probability from an initial Bloch state with a crystal momentum, $\mathbf{k}$ and energy, E to a final state, $\mathbf{k}'$. The method conclusively establishes the existence of Dirac fermions seen via linearly dispersing bands in the vicinity of the Dirac points. The experimental setup, ARPES data, and the hexagonal Brillouin zone (which we have discussed before) are shown in figure 3.15.

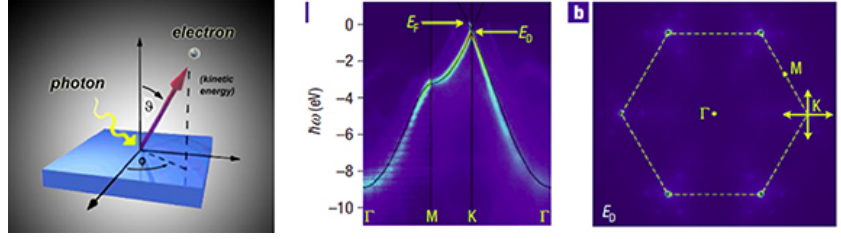


Figure 3.15. Plot showing the experimental setup for ARPES (left), the two linearly dispersing bands (middle) and the Dirac points are shown on the hexagonal BZ of graphene (right). Taken from reference [26], copyright 2006, by permission from Nature Physics: Springer Nature.

3.4.3 Landau levels in graphene

In order to proceed, we can write the Hamiltonian in a unified way that includes the description of both the Dirac points (valleys) $\mathbf{K}$ and $\mathbf{K}'$. So for each valley, we have one 2D spinor Hamiltonian. Thus augmenting the Hilbert space we can write the eigenfunctions as,

$$\psi = (\psi_{\mathbf{K}'}, \psi_{\mathbf{K}})^T \quad (3.13)$$

and the Hamiltonian is given by,

$$\mathcal{H} = \hbar v_F \begin{pmatrix} -\boldsymbol{\sigma}^* \cdot \mathbf{k} & 0 \\ 0 & \boldsymbol{\sigma} \cdot \mathbf{k} \end{pmatrix} = v_F \begin{pmatrix} -\boldsymbol{\sigma}^* \cdot \mathbf{p} & 0 \\ 0 & \boldsymbol{\sigma} \cdot \mathbf{p} \end{pmatrix}. \quad (3.13)$$

Now we shall be discussing the motion of the massless relativistic electrons in a magnetic field. In an usual 2D electron gas, the Landau quantization produces equidistant levels (see equation (3.37)), which is an artefact of the non-relativistic parabolic dispersion of the free carriers. We need to ascertain how the quantization formula is modified for the case of graphene.

As earlier, we do the Peierls substitution, $\mathbf{p} \rightarrow \mathbf{p} + e\mathbf{A}$. Thus the Hamiltonian becomes,

$$\mathcal{H}_{\mathbf{K},\mathbf{K}'} = v_F \begin{pmatrix} 0 & -(p_x + ip_y) & 0 & 0 \\ -(p_x - ip_y) & 0 & 0 & 0 \\ 0 & 0 & 0 & (p_x - ip_y) \\ 0 & 0 & (p_x + ip_y) & 0 \end{pmatrix}.$$

The wavefunction has now four components, namely,

$$\psi = \begin{pmatrix} \phi_A^{\mathbf{K}'} \\ \phi_B^{\mathbf{K}'} \\ \phi_A^{\mathbf{K}} \\ \phi_B^{\mathbf{K}} \end{pmatrix} \quad (3.13)$$

where $\phi_{A,B}^{\mathbf{K}}$ are the wavefunctions for an electron at momentum values corresponding to the valley $\mathbf{K}$ at the two sublattice sites A and B. Similar notations carry on for the other valley $\mathbf{K}'$. 2)

For a perpendicular magnetic field, $\mathbf{B} = B\hat{z}$, one can choose a Landau gauge, $\mathbf{A} = (-By, 0, 0)$. Since with this choice the Hamiltonian is independent of the spatial variable, x . So $[\mathcal{H}, p_x] = 0$ and hence p_x continues to be a good quantum number.

Further, the Hamiltonian in equation (3.132) is valley decoupled, that is, there are no matrix elements that connect the two valleys, namely, $\mathbf{K}$ and $\mathbf{K}'$. Thus it allows us look at the solutions at each valley separately. For the $\mathbf{K}$ point, we have a coupled equation for the wavefunctions, $\phi^{\mathbf{A}}$ and $\phi^{\mathbf{B}}$

$$\epsilon\phi_{\mathbf{A}}^{\mathbf{K}} = v_{\text{F}}(p_x - ip_y)\phi_{\mathbf{B}}^{\mathbf{K}} \quad (3.13)$$

$$\epsilon\phi_{\mathbf{B}}^{\mathbf{K}} = v_{\text{F}}(p_x + ip_y)\phi_{\mathbf{A}}^{\mathbf{K}}. \quad (3.13)$$

One can insert equation (3.133) in equations (3.134) and (3.134) in equation (3.133) to obtain, (4)

$$\epsilon^2\phi_{\mathbf{A}}^{\mathbf{K}} = v_{\text{F}}^2(p_x - ip_y)(p_x + ip_y)\phi_{\mathbf{A}}^{\mathbf{K}} \quad (3.13)$$

$$\epsilon^2\phi_{\mathbf{B}}^{\mathbf{K}} = v_{\text{F}}^2(p_x + ip_y)(p_x - ip_y)\phi_{\mathbf{B}}^{\mathbf{K}}. \quad (5)$$

Inserting the Landau gauge such that $p_x \rightarrow p_x + eBy$, (6)

$$\begin{aligned} \frac{\epsilon^2}{v_{\text{F}}^2}\phi_{\mathbf{B}}^{\mathbf{K}} &= (p_x + eBy + ip_y)(p_x + eBy - ip_y)\phi_{\mathbf{B}}^{\mathbf{K}} \\ &= \left[(p_x + eBy)^2 - i\{(p_x + eBy), p_y\} + p_y^2 \right] \phi_{\mathbf{B}}^{\mathbf{K}}. \end{aligned} \quad (3.13)$$

Since $[p_x, p_y] = 0$ and $[y, p_y] = i\hbar$, one gets, (7)

$$\frac{\epsilon^2}{v_{\text{F}}^2}\phi_{\mathbf{B}}^{\mathbf{K}} = [(p_x + eBy)^2 + e\hbar B + p_y^2]\phi_{\mathbf{B}}^{\mathbf{K}}. \quad (3.13)$$

Thus, we arrive at, (8)

$$\left(\frac{\epsilon^2}{v_{\text{F}}^2} - e\hbar B \right) \phi_{\mathbf{B}}^{\mathbf{K}} = (\tilde{p}_x^2 + \tilde{p}_y^2) \phi_{\mathbf{B}}^{\mathbf{K}} \quad (3.13)$$

where $\tilde{p}_x^2 = p_x + eBy$ and $\tilde{p}_y^2 = p_y$. Dividing both sides by $2m$, (9)

$$\frac{1}{2m} \left(\frac{\epsilon^2}{v_{\text{F}}^2} - e\hbar B \right) \phi_{\mathbf{B}}^{\mathbf{K}} = \frac{\tilde{p}_x^2 + \tilde{p}_y^2}{2m} \phi_{\mathbf{B}}^{\mathbf{K}} = \frac{1}{2} \tilde{k}(y - y_0)^2 + \frac{\tilde{p}_y^2}{2m} \phi_{\mathbf{B}}^{\mathbf{K}} \quad (3.14)$$

with $\tilde{k} = \frac{e^2 B^2}{m}$, $y_0 = \frac{p_x}{eB}$. Thus the RHS is identified as the Hamiltonian for a particle executing SHM in two dimensions about a coordinate point $(0, y_0)$. Thus it is obvious that the energy spectrum is given by, $\epsilon_n = (n + \frac{1}{2})\hbar\omega_B$ with $\omega_B = \frac{eB}{m}$. Hence, (10)

$$\frac{\epsilon^2}{v_{\text{F}}^2} = 2 \left(n + \frac{1}{2} \right) \hbar\omega_B - \hbar\omega_B = 2n\hbar\omega_B \quad \text{where } n = 0, 1, 2, \dots \quad (3.14)$$

equation (3.141) allows positive and negative roots for ϵ . So we can obtain the energy spectrum as, (1)

$$\epsilon = \text{sgn}(n) \sqrt{n} v_F (2\hbar e B)^{1/2}. \quad (3.14)$$

Let us define another quantity $\tilde{\omega} = v_F (2\hbar e B)^{1/2}$, so as to formally write the energy expression as that of a harmonic oscillator. We rewrite the above expression as,

$$\epsilon = \hbar \tilde{\omega} \text{sgn}(n) \sqrt{|n|}. \quad (3.14)$$

Thus, as opposed to the familiar harmonic oscillators when n can take positive integer values (including zero), however, here in graphene all integers, that is, both positive and negative numbers are allowed. The positive integers denote particles (or electrons) in the conduction band and the negative ones denote holes in the valence band. Further, unlike the 2DEG, here the Landau levels are not equidistant. The largest separation occurs between the lowest Landau level ($n = 0$) and the first one ($n = \pm 1$). This large gap essentially facilitates observation of QHE in graphene at large temperatures, which is even true for room temperature¹¹. The Landau levels for graphene are shown in figure 3.16. Clearly, they are non-equidistant, and hence quite distinct from those for the two-dimensional electron gas.

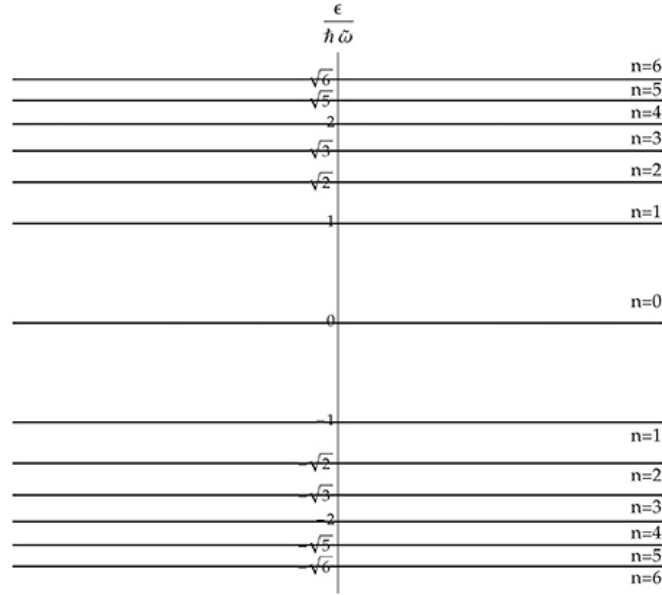


Figure 3.16. Plot showing the Landau levels in graphene for different indices, n .

So far we have been discussing spinless particles. Including the spin, there will be an additional two-fold degeneracy of the Landau levels owing to Zeeman splitting. A hierarchy of energy scales needs to be ascertained here. Let us compare the energy gap between the two lowest Landau levels and the Zeeman splitting corresponding to a typical magnetic field, B , for example, $B = 10$ T.

$$\Delta E_{LL} = \frac{\hbar \omega_B / 2}{v_F} \quad (\text{for the successive Landau levels}) \quad (3.14)$$

$$\Delta E_z = \sqrt{e \hbar B} \quad (\text{for the Zeeman term}). \quad (4)$$

For typical $v_F \simeq c/300$ (c : speed of light), (3.14 5)

$$\frac{\Delta E_z}{\Delta E_{LL}} \simeq 10^2. \quad (3.14 6)$$

Thus the Zeeman energy scale is much larger than the Landau level splitting, which makes it imperative to include spin degeneracy. Thus including the Zeeman term, the energy can be written as,

$$\frac{\epsilon^2}{v_F^2} = 2\hbar eB(n+1) \quad n = 0, 1, \dots \quad (3.14 7)$$

where the additional term in the RHS (denoted by $2\hbar eB$) accounts for the spin. Thus the energy level $\epsilon = 0$ is not present in the spectrum even for $n = 0$. This lowest Landau level is somewhat special in the following sense. The $n = 0$ level receives contribution from only one sublattice at each of the Dirac points. For example, ‘A’ sublattice contributes to the wavefunction at the Dirac point $\mathbf{K}$, and ‘B’ sublattice contributes at $\mathbf{K}'$. However, the $n \neq 0$ Landau levels have non-zero amplitudes at both the A and the B sublattices.

Finally, the wavefunctions corresponding to an arbitrary Landau level index at the two Dirac points are corresponding to the gauge we have chosen are,

$$\psi_{n,k}^{\mathbf{K}} = \frac{c_n}{\sqrt{L}} e^{-ikx} \begin{pmatrix} 0 \\ 0 \\ sgn(n)(-i)\phi_{|n|-1,k} \\ \phi_{|n|,k} \end{pmatrix} \quad (3.14 8)$$

and

$$\psi_{n,k}^{\mathbf{K}'} = \frac{c_n}{\sqrt{L}} e^{-ikx} \begin{pmatrix} \phi_{|n|,k} \\ sgn(n)(-i)\phi_{|n|-1,k} \\ 0 \\ 0 \end{pmatrix} \quad (3.14 9)$$

with

$$\begin{aligned} c_n(x) &= 1 & \text{for } n = 0 \\ &= \frac{1}{\sqrt{2}} & \text{for } n \neq 0 \end{aligned} \quad (3.15 0)$$

Further,

$$\begin{aligned} sgn(n) &= 0 & \text{for } n = 0 \\ &= \frac{n}{|n|} & \text{for } n \neq 0 \end{aligned} \quad (3.15 1)$$

with

$$\phi_{n,k} = \exp \left[-\frac{1}{2} \frac{(y - kl_B^2)^2}{l_B^2} \right] H_n \left[\frac{(y - kl_B^2)}{l_B} \right]$$

l_B is the magnetic length ($= \sqrt{\frac{\hbar}{eB}}$) as defined before, and $H_n(x)$ are the Hermite polynomials.^(3,15) $\phi_{n,k}$ denotes the eigenfunctions of an electron in presence of a magnetic field. n refers to the Landau level index. The Landau levels for different values of the flux, ϕ are shown in figure 3.17 as a function of k_x . As the magnitude of the flux is decreased, the width of the flat band appearing at the Fermi energy (at $E/t = 0$) decreases. Further, the flat bands become dissipative in the bulk corresponding to larger values of the Landau level index, n and lower values of the flux ϕ . For a weak magnetic field, such that, $\phi/\Phi_0 = 1/1600$, the energy bands of the bulk regain their Dirac-like structure similar to the single-particle energy levels, while the zero mode flat band continues to exist.

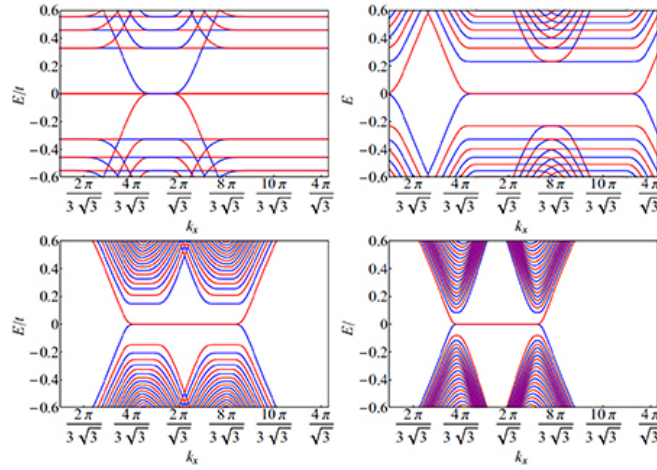


Figure 3.17. Plot showing the Landau levels in graphene for different values of flux, ϕ . The values of the fluxes are $\phi = \frac{\Phi_0}{100}$ (for left upper panel), $\phi = \frac{\Phi_0}{200}$ (for right upper panel), $\phi = \frac{\Phi_0}{500}$ (for left lower panel) and $\phi = \frac{\Phi_0}{1600}$ (for right lower panel). Here $\Phi_0 = \frac{h}{e}$ is the flux quantum.

The experimental results for the (room temperature) QHE in graphene is shown in figure 3.18. The Hall conductivity as a function of the gate voltage, V_g shows plateaus at $2e^2/h$, and the longitudinal resistivity (ρ_{xx}) vanishes ($< 10\Omega$) at values of the voltage where the Hall conductivity saturates to values that are multiples of e^2/h . The maximum difference between the non-equidistant Landau levels is about $\Delta E \simeq 600 \text{ K}^{12}$ at $B = 29 \text{ T}$ and survival of QHE at very large temperature is owing to a large cyclotron frequency, ω_c . At still larger values of the magnetic field, namely, $B = 45 \text{ T}$, ΔE is about 2800 K when the Fermi energy lies between the lowest Landau level ($n = 0$) and the first Landau levels ($n = \pm 1$). Thus at this value of the magnetic field, the energy scale $\hbar\omega_c$ exceeds the thermal energy at $T = 300 \text{ K}$ by an order of magnitude. The density of the carriers in graphene is very high ($\sim 10^{13} \text{ cm}^{-2}$) such that the lowest Landau level is fully populated even at very large values of the field.

We have included numeric computation of the Hall conductivity as a function of the Fermi energy at comparable values (to those in experiments) of the magnetic field, namely, $B = 30 \text{ T}$ and 50 T using Kubo formula in figure 3.18. The quantization of the Hall conductivity is clearly visible. Further, the longitudinal conductivity, σ_{xx} is shown for one of them, namely, 50 T . σ_{xx} shows vanishingly small values corresponding to plateaus of the Hall conductivities, while it shows spikes when the Hall conductivity jumps from one plateau to another.

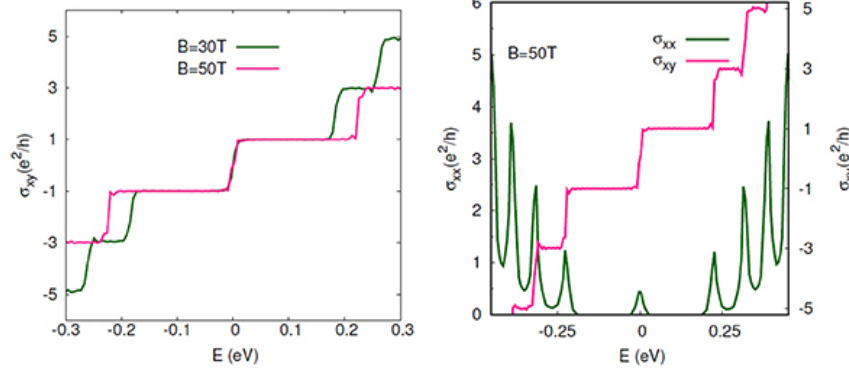


Figure 3.18. Plot showing the variation of Hall conductivity as a function of the bias voltage for a particular value of the magnetic field, such as, $B = 30$ T and 50 T. The plateaus in the Hall conductivity are clearly visible. The longitudinal conductivity (σ_{xx}) is shown for a specific value, namely, $B = 50$ T.

3.4.4 Experimental observation of the Landau levels in graphene

There are primarily two experimental techniques for observing the existence of Landau levels. They are (i) the infra-red (IR) spectroscopy [24] and (ii) scanning tunnelling microscopy (STM) [25] experiments. In the following we include a brief discussion on each of them and their utilities in observing the non-equidistant Landau levels in graphene.

In the IR spectroscopy, the optical transitions from one Landau level to another are studied via measuring the cyclotron frequencies. The Landau levels being proportional to $\sqrt{n}$ (n : Landau level index), all the frequencies of the optical transitions are distinct as the energy spacing between each pair of Landau level is different than the other pairs. These optical transitions are of two types which correspond to transitions between the electron or the hole states in the conduction or the valence bands (intra-band transition), or the transitions between the electron and the hole states pertaining to the valence and conduction bands (inter-band transition), respectively. The photoconductive response and the resistive voltage show the existence of differently spaced Landau levels in graphene at particular values of the magnetic field, longitudinal current, and the IR frequency. The photoconductive intensity as a function of the carrier density, n (not to be confused with the Landau level index) show distinct peaks which are proportional to the energy absorbed from the incident IR radiation.

In the STM experiment, the specific energy levels can be identified by varying the bias voltage between the tip and the surface of the sample, and the tunnelling current generated is proportional to the local density of states. In graphene, the Landau levels are directly observed via the peaks in the tunnelling spectrum. From the positions of the peaks as a function of the sample bias (shown in figure 3.20), the energies of the Landau levels can be extracted. In the STM spectra shown below corresponding to a specific temperature (4.4 K) and magnetic field in the range $[0, 12$ T], the peaks occurring at the zero sample bias (marked as ‘A’ peak) do not suffer a change in position as the magnetic field is ramped up from 0 to 12 T, while the peaks on either side of the zero bias (marked as ‘Z’ peak) show a gradual fanning outward with increase in the applied magnetic field. Clearly, the ‘A’ peaks correspond to the $n = 0$ Landau level, while the ‘Z’ peaks correspond to $n = \pm 1$ levels. Closer analysis of the other peaks reveals a $\sqrt{n}$ dependence. In the lower left panel of figure 3.20 the STM spectra are shown as a function of $E/\sqrt{B}$. Further, the peak positions conform to a $\sqrt{B}$ dependence where the Landau level indices are

shown by different peaks as a function of a scaled energy variable, namely, $E/\sqrt{B}$. In the other figure (right panel of figure 3.20), $E/\sqrt{B}$ is shown as a function of the Landau level index, n and also goes as $\text{sgn}(n)\sqrt{|n|}$ where the latter shows a linear behaviour passing through the origin (right bottom lower panel of figure 3.19).

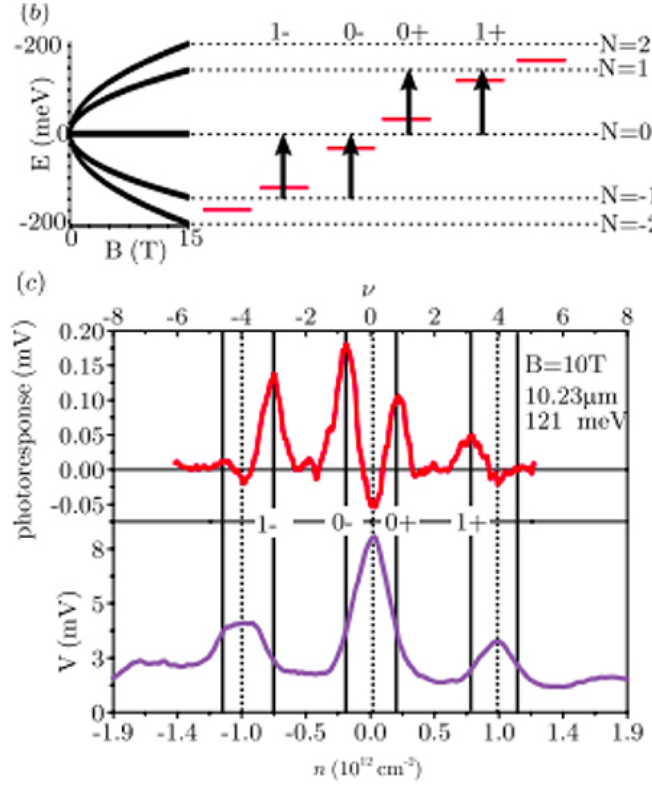


Figure 3.19. Plot showing the identification of distinct Landau levels which go as $\sqrt{n}$ (reprinted with permission from reference [13], copyright (2007) by the American Physical Society).

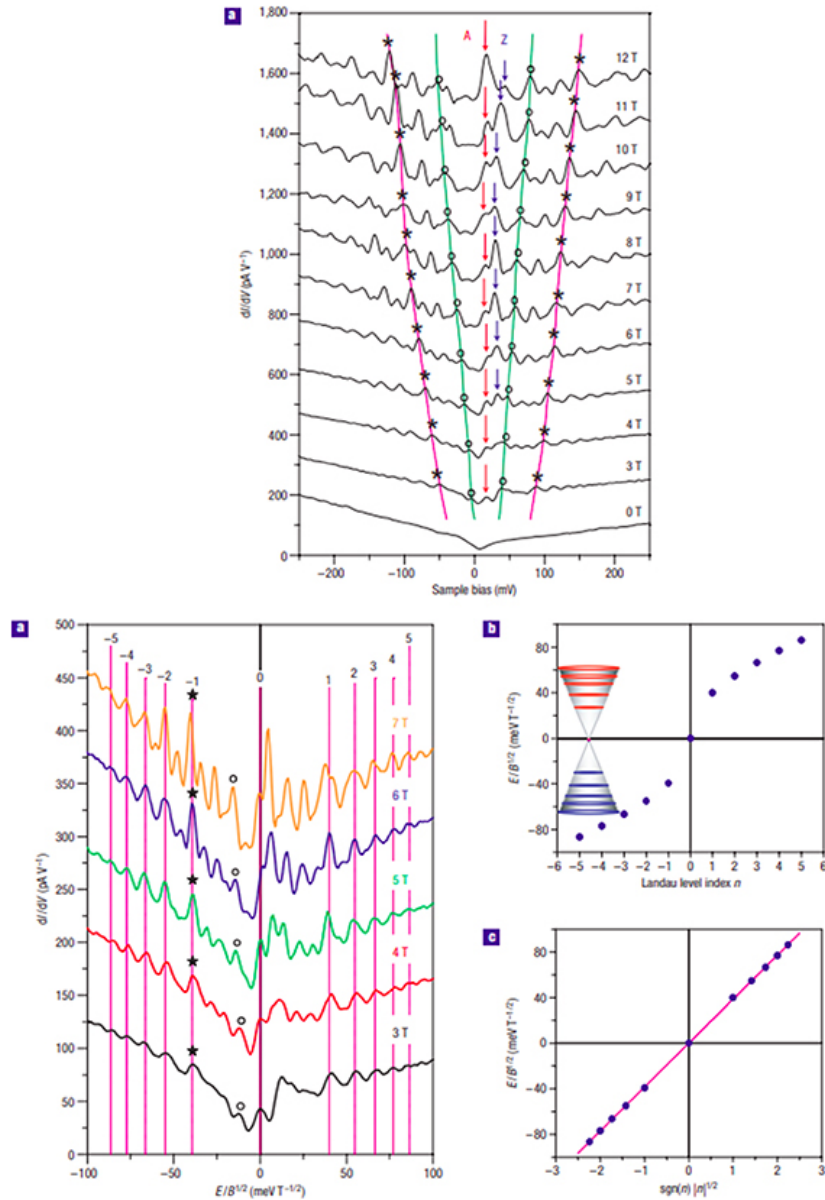


Figure 3.20. Plot of the tunnelling spectra showing Landau levels which go as $\sqrt{n}$ (taken from reference [25], copyright 2007 by permission from Nature Physics: Springer Nature).

3.4.5 Summary

We begin with a historical overview of the QHE. The experiment and the physical systems are described with an emphasis on the 2D nature of the ‘dirty’ electronic system in the presence of a strong perpendicular magnetic field at low temperature. The Hall resistivity as a function of the field shows quantized plateaus in unit of h/e^2 with an accuracy of one part in more than a billion. Very surprisingly, the longitudinal resistivity synergetically vanishes at the positions of the plateaus for the Hall resistivity. This indicates emergence of a phase with an inherent ambiguousness of being a perfect conductor and a

perfect insulator at the same time. However, such an ambiguity can only be reconciled for an electron gas confined in a plane in the presence of a magnetic field.

Quite intriguingly, the presence of the perpendicular magnetic field introduces ‘another’ quantization, which replaces the band structure (energy as a function of the wavevector) of the electronic system. This quantization was shown via solving the Schrödinger equation in the presence of a Landau gauge. The resultant energy levels of this problem are the infinitely degenerate Landau levels, which slightly broaden due to the presence of impurity and disorder, but still remain distinct and cause quantization of the Hall conductivity as the magnetic field is ramped up gradually. Further, this quantization is visioned as a quantum pump by Laughlin where an electron gas in a planar disk geometry subjected to a magnetic field shows a transfer of one unit of charge from the inner to the outer edge of the disk as the magnetic flux changes by one quantum ($= \frac{h}{e}$). Further, we have studied the same problem in the circular gauge which is relevant to the study of FQHE.

The quantum Hall state is the first realization of a topological insulator where the transmission of charges occurs through the edges of the electronic system, while the bulk remains insulating. The quantization of the Hall conductivity¹³ is shown to be a topological invariant called the Chern number, which can only assume integer values. We have further derived the Kubo formula to compute the Hall conductivity.

Finally, we have chosen graphene as another hobby horse to demonstrate occurrence of QHE. The choice of graphene stems from the fact that the energy scale of the problem allows QHE to be realizable at temperatures as high as room temperature, or even higher than that. The Landau levels of graphene are computed and are shown to be distinct from those for the 2DEG. Lastly, the experimental demonstration and observation of QHE in graphene are discussed.

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¹Parabolic dispersion in 3D yields DOS as $E^{1/2}$ (E : energy), in 2D it is constant, while the DOS varies as $E^{-1/2}$ in 1D.

²Which should be valid for weak magnetic fields, and is not exactly true for QHE, but nevertheless it serves our purpose.

³Remember that $\mathbf{E} = 0$ in the lab frame, though $\mathbf{B}$ remains unchanged.

⁴A combination of the two yields a ‘symmetric’ gauge which we shall introduce and employ later.

⁵The value of $\phi_0 \simeq 4.13 \times 10^{-15}$ Wb-m².

⁶ $(y + \frac{\partial}{\partial y})u(y) = 0$, or $\frac{\partial u}{u} = -ydy$. The solution is $u = e^{-y^2/2}$.

⁷In fact, naively it seems ironical that a broken time reversal symmetry is solely responsible for quantization of the Hall plateaus.

⁸ $\Phi_i \bmod \Phi_0$ refers to the remainder when Φ_i is divided by Φ_0 .

⁹It should be remembered that ϕ is not a measurable quantity, while $e^{i\phi}$ is a measurable quantity.

¹⁰The Dirac equation is written in conventional notations as, $\mathcal{H} = c\boldsymbol{\alpha} \cdot \mathbf{p} + \beta mc^2$ where α and β are Hermitian operators which do not operate on the space and time variables. In the case of graphene, the second term is absent.

¹¹The title of the paper by Geim and Novoselov is ‘*Room temperature quantum Hall effect in graphene*’. See [23].

¹² ΔE is called the activation energy in the original paper by [23].

¹³It is more appropriate to talk about conductivity, rather than the resistivity.